Wavelet-Driven Spectral–Spatial State-Space Modeling for Hyperspectral Image Classification

Sihao Zeng, Zhen Wang, *Member, IEEE*, Jiayuan Li, Ruixiang Li, and Zhuhong You, *Member, IEEE*

***Abstract*—Hyperspectral image (HSI) classification is challenged by high spectral dimensionality and complex spectral–spatial dependencies. Existing models, including convolutional neural networks (CNNs), Transformers, and Mamba, struggle to balance long-range modeling, computational efficiency, and preservation of spectral–spatial coherence. To address these limitations, we propose the wavelet-driven spectral–spatial state space network (WD4SNet), a novel and efficient architecture for HSI classification. Specifically, WD4SNet first employs a spectral–spatial fusion block (SSFB) to perform complementary integration of local spatial textures and spectral information at shallow layers, providing robust initial representations. The core of the network is a spectral–spatial state-space model (SSM), which consists of a spectral-based multiscale state space (SMSS) and a spatial-axis multiscale state space (SASS). Both modules incorporate wavelet-guided enhancement with learnable subband processing, enabling multiscale global and local spectral–spatial dependency modeling with linear complexity. To further enhance spectral–spatial synergy, a cross-axis spectral–spatial feature alignment (CASFA) module is introduced for adaptive alignment and fusion of spectral and spatial representations. Finally, a lightweight feature pyramid decoder (LFPD) reconstructs high-resolution prediction maps with minimal computational overhead. Extensive experiments on four benchmark hyperspectral datasets demonstrate that WD4SNet consistently outperforms state-of-the-art methods across multiple quantitative evaluation metrics. Furthermore, ablation studies validate the effectiveness of each module in advancing spectral–spatial fusion and representation learning. All codes and classification results will be available on https://github.com/darkseid-arch/WD4SNet**

***Index Terms*—Feature alignment, hyperspectral classification, pyramid decoder, spectral–spatial modeling, state-space model (SSM).**

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I. Introduction

HYPERSPECTRAL image (HSI) acquires hundreds of contiguous spectral bands across the visible to far-infrared spectrum, enabling the construction of high-dimensional data representations that combine fine-grained spectral signatures with rich spatial information [1], [2], [3]. This capability allows for precise discrimination of spectrally similar materials and has established HSI as an essential technology in diverse domains, including geological exploration [4], precision agriculture [5], urban planning [6], and environmental monitoring [7]. Hyperspectral image classification, the task of assigning each pixel to a predefined land-cover category, directly determines the practical utility of HSI in real-world applications and thus remains a central research focus in remote sensing. However, the intrinsically high dimensionality of HSI data and the significant spectral similarity among different materials [8] continue to pose substantial challenges for achieving accurate and robust classification.

To address these challenges, HSI classification methods have advanced along two principal directions: traditional machine learning and modern deep learning. Conventional approaches, such as support vector machines (SVMs) [9] and k-nearest neighbors (KNNs) [10], primarily exploit spectral information but are limited by poor generalization and reliance on handcrafted features in high-dimensional settings. In contrast, deep learning methods, with their powerful automatic feature extraction capabilities, have become the mainstream paradigm. Representative architectures include convolutional neural networks (CNNs), Transformers, and the Mamba framework. CNNs leverage 2-D/3-D or hybrid convolutions (e.g., HybridSN [11]) to capture spatial–spectral features; Transformers (e.g., SpectralFormer [12]) utilize global self-attention to model long-range dependencies; while Mamba-based models (e.g., MambaHSI [13]) employ linear-complexity state space mechanisms for efficient long-sequence spectral modeling.

In addition to single-source HSI classification, recent studies have also explored multisource remote sensing data classification by integrating HSI with SAR or LiDAR data. For example, the representative spectral correlation network (RSCNet [14]) addresses spectral redundancy and multisource heterogeneity by adaptively selecting task-relevant spectral bands under cross-source guidance and enhancing feature interaction through cross-source adaptive fusion.

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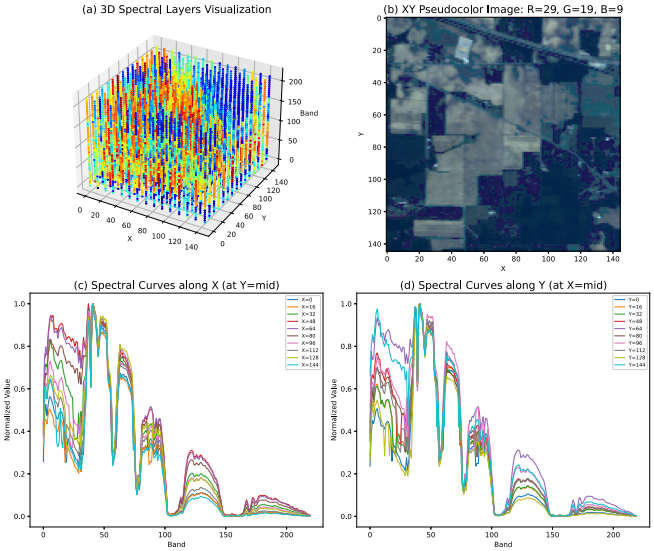


Fig. 1. Illustration of the spectral and spatial characteristics of HSI. (a) 3-D visualization of spectral layers. (b) Pseudo-color image. (c) Spectral curves along the X-axis. (d) Spectral curves along the Y-axis.

The prototype-based information compensation network (PICNet [15]) further improves multisource land-cover classification by introducing frequency interaction and prototype-based information compensation to model global complementary information across modalities. These methods demonstrate the effectiveness of exploiting multisource complementary cues for remote sensing classification. However, they mainly rely on additional SAR or LiDAR data, while the present work focuses on single-source HSI classification and aims to enhance the internal spectral–spatial representation of HSI through wavelet-driven state-space modeling.

Despite recent advances, existing HSI classification models exhibit notable limitations: CNNs are constrained in capturing long-range spatial dependencies, Transformers incur prohibitive computational costs due to quadratic complexity, and Mamba-based models may disrupt local spatial structures and insufficiently exploit spectral context. As illustrated in Fig. 1, these challenges are manifested as follows.

1) *Spectral-Redundant Overhead:* The intrinsic high dimensionality and spectral redundancy of HSI data are evident in the 3-D spectral visualization [see Fig. 1(a)], where densely sampled spectral bands lead to substantial information redundancy and elevated computational overhead for vision-oriented models.
2) *Spatial Structure Disruption:* The pseudo-color image in Fig. 1(b) highlights the complex spatial organization and heterogeneous distribution of ground objects. However, unidirectional or simplistic spatial modeling often disrupts local pixel adjacency, resulting in blurred class boundaries and degraded spatial precision.
3) *Insufficient Spectral Context Modeling:* The spectral curves along the X- and Y-axes [see Fig. 1(c) and (d)] demonstrate significant interband correlations and contextual dependencies. Existing unidirectional sequence models inadequately capture these relationships, leading to suboptimal exploitation of spectral information and loss of discriminative power.

To address these challenges, we propose the wavelet-driven spectral–spatial state space network (WD4SNet), an efficient architecture that integrates multiscale wavelet processing with state-space modeling for hyperspectral image classification. WD4SNet begins with a spectral–spatial fusion block (SSFB) that adaptively combines spatial textures and spectral signatures at shallow layers for robust feature initialization. The core spectral–spatial state-space model (SSM) comprises two parallel branches: a spectral-based multiscale state space (SMSS) and a spatial-axis multiscale state space (SASS), both enhanced by wavelet-guided enhancement with learnable subband processing to capture spectral and spatial dependencies at multiple scales with linear complexity. A cross-axis spectral–spatial feature alignment (CASFA) module is then employed to explicitly align and integrate spectral and spatial features via bidirectional attention and gating mechanisms. Finally, a lightweight feature pyramid decoder (LFPD) aggregates multiscale features to produce high-resolution predictions with minimal computational cost. WD4SNet is not a simple combination of these components. Its main motivation is to redesign and integrate them according to the intrinsic characteristics of hyperspectral images, including spectral redundancy, spectral continuity, complex spatial structures, and efficient long-range dependency modeling. Specifically, SSFB introduces weighted convolution to enhance local spectral–spatial continuity, SMSS and SASS adopt a global–local–identity strategy to separately model long-range dependencies, local variations, and stable information flow, and CASFA performs cross-axis feature alignment before fusion. Therefore, WD4SNet provides an HSI-oriented collaborative modeling framework rather than a straightforward stacking of existing modules. The main contributions of this work are summarized as follows.

1) We propose WD4SNet, an HSI-oriented spectral–spatial collaborative modeling framework that integrates weighted shallow feature fusion, axis-wise state-space modeling, cross-axis feature alignment, and lightweight multiscale decoding.
2) We design SSFB with weighted 2-D/3-D convolutions to better exploit local spatial correlation and spectral continuity in hyperspectral images.
3) We develop SMSS and SASS with a global–local–identity divide-and-conquer strategy along the spectral and spatial axes, enabling the model to jointly capture long-range dependencies, local fine-grained variations, and stable residual information.
4) We introduce CASFA to align spectral and spatial representations through bidirectional attention and adaptive channel-wise gating, rather than using simple addition, concatenation, or parallel fusion.

The article is structured as follows. Section II provides an overview of the related works. Section III introduces the proposed WD4SNet framework and its associated components. The experimental results and discussion analysis are presented in Section IV. Finally, the conclusion is drawn in Section V.

II. Related Works

A. Traditional Methods for HSI Classification

Early HSI classification predominantly employed traditional machine learning algorithms, where discriminative features were extracted via handcrafted design or statistical modeling. Representative approaches include SVMs [9], KNNs [10], decision trees [16], and logistic regression (LR) [17]. SVM, in particular, was widely adopted for its ability to cope with high-dimensional and small-sample scenarios through kernel methods. However, these techniques exhibit two principal limitations: 1) their generalization performance degrades rapidly in high-dimensional HSI spaces, often succumbing to the curse of dimensionality; and 2) they rely heavily on handcrafted features, necessitating expert-driven design of spectral indices or texture descriptors, which restricts adaptability to complex and heterogeneous land-cover scenes. To address these challenges, representation learning methods such as low-rank representation [18] and sparse representation [19] were introduced to exploit interband correlations for dimensionality reduction. While these methods offer improved feature extraction, they remain fundamentally dependent on manual expertise and struggle to discriminate spectrally similar or spatially diverse materials. As a result, traditional methods have been largely superseded by deep learning models capable of automatic feature learning.

B. CNNs-Based HSI Classification Methods

The advent of deep learning has established CNNs as a cornerstone for HSI classification, owing to their capacity for automatic feature extraction and local spatial modeling. Early explorations, such as the five-layer hierarchical CNNs by Hu et al. [20], demonstrated the feasibility of leveraging deep architectures for spectral feature learning in HSIs. To further exploit spatial dependencies, Makantasis et al. [21] proposed a hybrid model combining 2D-CNNs with multilayer perceptrons (MLPs), while Fang et al. [22] introduced the stacked multibias network (SMBN) to improve spatial discrimination via adaptive response selection. For joint spectral–spatial representation, 3D-CNNs are adopted by Hamida et al. [23], enabling direct modeling of HSI cubes. Zhong et al. [24] advanced this approach with the spectral–spatial residual network (SSRN), employing multiple 3-D convolutions for enhanced feature integration. Hybrid architectures, such as HybridSN [11], combine 3-D convolutions for spectral–spatial modeling with 2-D convolutions for efficient local detail capture. To alleviate optimization difficulties in deeper networks, residual [25] and dense [26] connections have been incorporated to facilitate gradient flow and stable training. SMESC [27] further improves spectral and spatial modeling by integrating subpixel feature amplification and channel modulation. In addition to pure CNN architectures, recurrent and attention-enhanced convolutional models have also been explored for spectral–spatial classification. ASSMN [28] combines LSTM-based spectral modeling with ConvLSTM-based spatial modeling to extract adaptive multiscale contextual features. A2S2K [29] further introduces attention-based adaptive spectral–spatial kernels to selectively enhance discriminative spectral–spatial responses. These methods improve local spectral–spatial representation, but their convolutional or recurrent structures still have limitations in modeling long-range dependencies efficiently.

C. Transformer-Based HSI Classification Methods

To address the inherent locality of CNNs, Transformer architectures equipped with attention mechanisms have been introduced to HSI classification, establishing a new paradigm for spectral–spatial representation learning. Transformers excel at modeling long-range dependencies, enabling comprehensive contextual understanding across both spectral and spatial domains. For instance, He et al. [30] proposed HSI-BERT, employing multihead self-attention to capture broad contextual relationships and mitigate spatial distance constraints. Recognizing the sequential nature of spectral bands, Hong et al. [12] developed SpectralFormer, which constructs grouped spectral embeddings to enhance spectral feature discrimination. Sun et al. [31] introduced the spectral–spatial feature tokenization Transformer, integrating a Gaussian-weighted tokenization to jointly capture semantic and spectral–spatial characteristics. Recent Transformer-based HSI classification methods further improve spectral–spatial representation through local spatial recovery and memory-enhanced feature interaction. HiT [32] introduces convolutional projections and permutation-based operations to enhance fine-grained spectral–spatial modeling. SPRLT [33] employs local Transformer modeling with spatial partition recovery to reduce spatial information loss, while MASSFormer [34] introduces memory tokens to enhance deep spectral–spatial feature interaction. Morphformer [35] further combines morphological convolutions with attention mechanisms to facilitate robust feature interactions. Despite their advances, Transformers are fundamentally limited by the quadratic computational complexity of self-attention, resulting in significant parameter redundancy and memory overhead for high-dimensional HSIs. To alleviate this, various lightweight architectures have been proposed: GSC-ViT [36] employs grouped separable convolutions to reduce computational cost; MMAN [37] integrates multiscale CNNs with attention for hierarchical modeling; CASST [38] and HMSSF [39] focus on cross-attention and multiscale integration to improve spectral–spatial feature fusion. Transformers overcome the local receptive field constraints of CNNs and excel at capturing long-range dependencies. However, their scalability is severely hampered by quadratic complexity, making them less practical for large-scale or high-dimensional HSI classification.

D. Mamba-Based HSI Classification Methods

Mamba architectures based on structured SSMs have emerged as efficient alternatives for HSI classification, offering linear computational complexity and effective long-range sequence modeling. The foundation of this paradigm was established by Gu et al. [40], which introduced SSMs for linear-complexity sequence modeling, and was further advanced by Mamba [41] through a selective state mechanism for enhanced contextual compression. Leveraging these

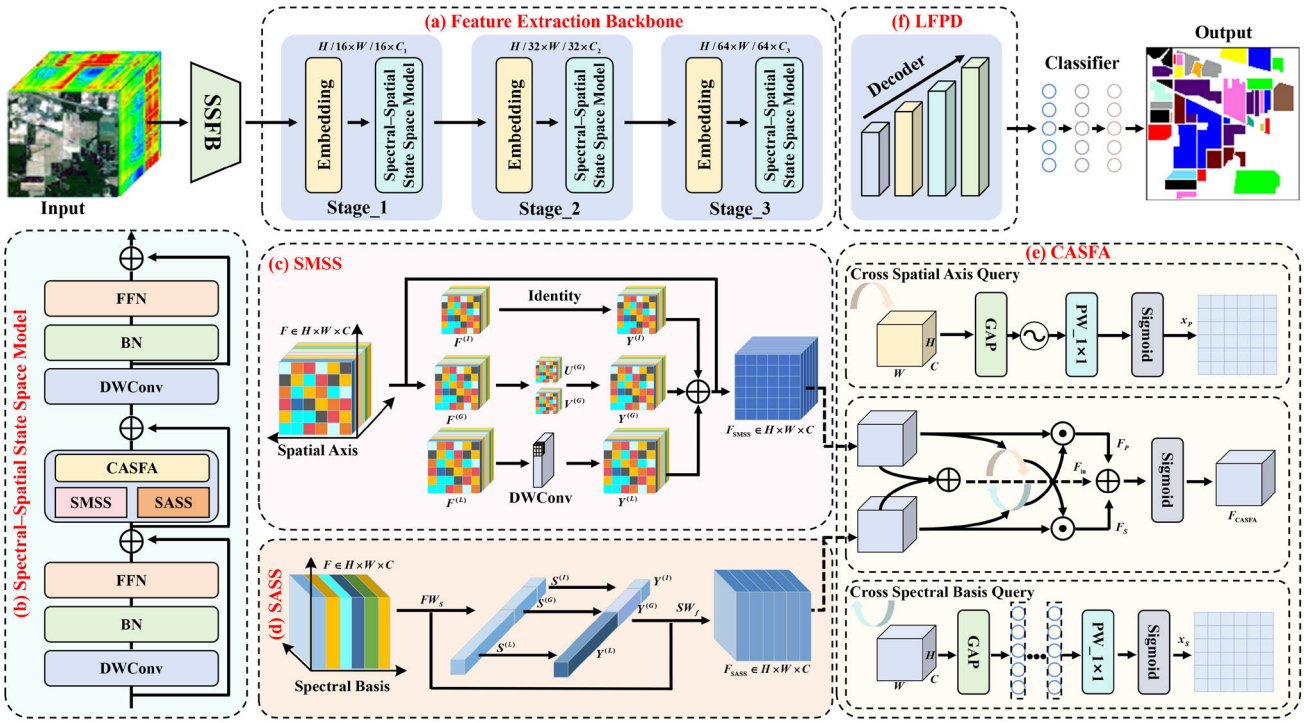


Fig. 2. Overall architecture of WD4SNet. The pipeline comprises: an SSFB that adaptively integrates spectral and spatial information; (a) feature extraction backbone, where each stage includes patch embedding and (b) S4M featuring parallel (c) SMSS and (d) SASS branches; (e) CASFA for explicit feature fusion; and (f) LFPD that aggregates multiscale features to generate dense prediction maps.

properties, Mamba-based architectures have demonstrated superior scalability and performance compared to conventional Transformer models, particularly aligning with the spectral continuity characteristic of HSI data. In HSI classification, MambaHSI [13] employs Mamba blocks to jointly capture long-range spatial and spectral dependencies, addressing the inefficiencies of Transformer-based spectral–spatial fusion. To further improve efficiency, He et al. [42] proposed the interval group spectral–spatial Mamba, which utilizes grouped interval modules for multidirectional spectral–spatial information extraction. SSMamba [43] introduces a spectral–spatial token generation mechanism with stacked Mamba blocks to enhance spectral–spatial feature integration. Zhou et al. [44] developed Mamba-in-Mamba (MiM), employing centralized cross-scanning to mitigate local information loss from unidirectional scanning. DBMGNet [45] combines a Mamba branch with a graph convolution branch to jointly capture long-range dependencies and non-Euclidean spatial relationships. FAHM [46] incorporates frequency-aware hierarchical modeling and bidirectional adaptive Mamba blocks to improve robust spectral–spatial representation. These methods demonstrate the potential of Mamba for HSI classification. In addition, vision-oriented variants such as ViM [47] and Vmamba [48] extend Mamba to visual recognition via multidirectional spatial scanning; despite their advantages, current Mamba-based models face persistent challenges. Unidirectional spatial scanning can disrupt local adjacency, leading to blurred boundaries; one-way spectral information flow limits interband context exploitation, resulting in information loss; and vision-specific variants do not address HSI-specific dimensionality and redundancy, incurring unnecessary computation. These limitations underscore the need for further architectural innovations tailored to the distinct properties of hyperspectral data.

III. Methodology

A. Overall Structure

The overall architecture of WD4SNet for hyperspectral image classification is illustrated in Fig. 2. Given an input HSI $\boldsymbol{X} \in \mathbb{R}^{B\times C\times H\times W}$, the model first employs an SSFB to directly process $\boldsymbol{X}$. The SSFB performs channel-wise linear recombination and weighted convolution, producing initial representations $\boldsymbol{F}_{\text{SSFB}} \in \mathbb{R}^{B\times C_1\times H\times W}$ that adaptively fuse spatial textures and spectral continuity while preserving spatial resolution. These fused features are then passed through a patch embedding, which reshapes and downsamples the representations to $\boldsymbol{F}_0 \in \mathbb{R}^{B\times C_1\times H/s\times W/s}$, where s is the downsampling factor. The resulting tokens serve as input to a multistage feature extraction backbone. At each stage l ($l = 1, 2, 3$), features $\boldsymbol{F}^{(l-1)}$ are processed by a spectral–spatial SSM (S4M), which decomposes information along spectral and spatial axes via two parallel branches: the SMSS and the SASS. Specifically, SMSS projects $\boldsymbol{F}^{(l-1)}$ to a spectral basis and applies wavelet-enhanced state-space modeling to capture long-range spectral dependencies, generating $\boldsymbol{F}^{(l)}_{\text{SMSS}}$; SASS exploits bidirectional Mamba and multikernel convolutions to capture spatial context, yielding $\boldsymbol{F}^{(l)}_{\text{SASS}}$. The outputs $\boldsymbol{F}^{(l)}_{\text{SMSS}}$ and $\boldsymbol{F}^{(l)}_{\text{SASS}}$ are then adaptively aligned and fused by the CASFA, resulting in $\boldsymbol{F}^{(l)}_{\text{CASFA}}$. Feature maps from all stages, $\{\boldsymbol{F}^{(1)}_{\text{CASFA}}, \boldsymbol{F}^{(2)}_{\text{CASFA}}, \boldsymbol{F}^{(3)}_{\text{CASFA}}\}$, are aggregated by an LFPD to

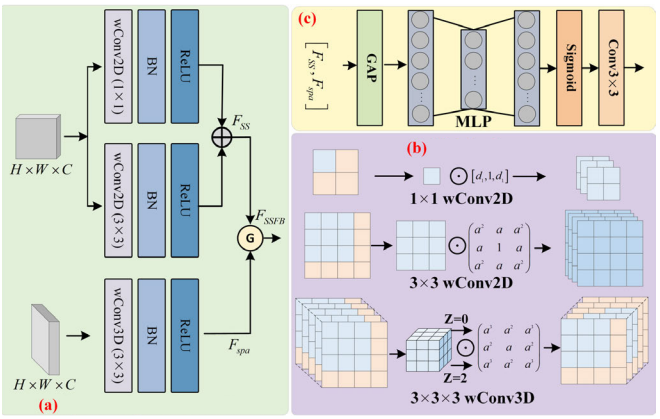


Fig. 3. Architecture of the SSFB, featuring parallel spatial and spectral–spatial branches with weighted convolutions and an adaptive channel-wise gating fusion mechanism. (a) SSFB. (b) Weight conv. (c) Gate.

recover multiscale contextual information and spatial details. The final dense prediction map is generated by a classification head, producing $\boldsymbol{Z} \in \mathbb{R}^{B\times C_{\text{cls}}\times H\times W}$, where $\boldsymbol{C}_{\text{cls}}$ is the number of classes. The proposed spectral–spatial decomposition is motivated by the heterogeneous nature of HSI data. Spectral features mainly represent continuous material responses across bands, while spatial features describe local textures, structures, and object boundaries. Directly fusing them through addition, concatenation, or a single unified branch may mix heterogeneous cues too early and weaken either spectral discrimination or spatial structure preservation. Therefore, WD4SNet first models spectral and spatial dependencies along their corresponding axes and then performs adaptive alignment and fusion. This decomposition provides a clearer division of modeling responsibilities and is more suitable for preserving both spectral continuity and spatial consistency.

B. Spectral–Spatial Fusion Block

The SSFB is designed to perform robust fusion of shallow spatial textures and spectral characteristics, providing discriminative and noise-resilient features for subsequent processing. As illustrated in Fig. 3, the SSFB consists of two parallel branches: a spatial branch and a spectral–spatial joint branch, whose outputs are adaptively fused by a channel-wise gating mechanism.

Given input features $\boldsymbol{X} \in \mathbb{R}^{B\times C\times H\times W}$, the spatial branch aims to enhance local structure and interband interaction while preserving spatial resolution.

First, a channel-wise linear recombination is performed via a 1×1 convolution

$$\boldsymbol{F}_{b,c}^{(1\times 1)}(x,y) = \sum_{c'} W_{c,c'}^{(1\times 1)} X_{b,c'}(x,y) \tag{1}$$

where $W^{(1\times 1)} \in \mathbb{R}^{C\times C}$ is a learnable weight matrix. A symmetric nonnegative weighting vector $\alpha = [d_1, \ldots, d_m, 1, d_m, \ldots, d_1]$ is constructed by softplus-transformed parameters d_i, and the 2-D template is defined as $\Phi^{2d} = \alpha\alpha^T$, where $\Phi^{2d} \in \mathbb{R}^{k\times k}$, typically $k = 3$. This template modulates a 3×3 convolution kernel $W^{(3\times 3)}$ by element-wise multiplication. The enhanced spatial features are then calculated as

$$\boldsymbol{F}_{\text{spa}}(x,y) = \sum_{c'} \sum_{u,v=-1}^{1} \left(W^{(3\times 3)} \odot \Phi^{2d}\right) \boldsymbol{F}_{b,c'}^{(1\times 1)} \tag{2}$$

where $\odot$ denotes element-wise multiplication. The spectral–spatial joint branch captures both spectral continuity and local spatial correlation by applying a 3-D weighted convolution to spectral–spatial cubes. The 3-D template is formed as $\Phi^{3d} = \alpha \otimes \alpha \otimes \alpha \in \mathbb{R}^{k\times k\times k}$, where $\otimes$ denotes the outer product. The weighted 3-D kernel is then

$$\tilde{W} = W^{(3\times 3\times 3)} \odot \Phi^{3d} \tag{3}$$

where $W^{(3\times 3\times 3)}$ is the learnable 3-D convolution kernel. The spectral–spatial features are obtained by

$$\boldsymbol{F}_{\text{ss}}(z,x,y) = \sum_{c'} \sum_{p,q,r=-1}^{1} \tilde{W}_{c,c'}(p,q,r)\, X_{b,c'} \tag{4}$$

where (p, q, r) represent spectral and spatial offsets. The outputs of both branches, F_{spa} and $F_{\text{ss}} \in \mathbb{R}^{B\times C\times H\times W}$, are fused by a channel-wise gating mechanism. The features are concatenated along the channel dimension and globally pooled, followed by an MLP and sigmoid activation to produce gating coefficients

$$\gamma = \sigma\left(\text{MLP}\left(\text{GAP}\left(\left[\boldsymbol{F}_{\text{spa}}, \boldsymbol{F}_{\text{ss}}\right]\right)\right)\right) \tag{5}$$

where GAP denotes global average pooling, $[\cdot,\cdot]$ is channel concatenation, and $\sigma(\cdot)$ is the sigmoid function. The final fused output is as follows:

$$\boldsymbol{F}_{\text{SSFB}} = \gamma \odot \boldsymbol{F}_{\text{spa}} + (1-\gamma) \odot \boldsymbol{F}_{\text{ss}} \tag{6}$$

where $\gamma \in [0,1]^{B\times C\times 1\times 1}$. The enhanced feature $\boldsymbol{F}_{\text{SSFB}}$ is then projected to the required embedding dimension and downsampled for the next stage via a stride-based 3×3 convolution with batch normalization and activation

$$\boldsymbol{F}_i = \text{BN}\left(\text{Conv}_{3\times 3,s}\left(\boldsymbol{F}_{\text{SSFB}}\right)\right), \quad \boldsymbol{F}_{\text{embed}} = \Phi\left(\boldsymbol{F}_i\right) \tag{7}$$

where s is the stride, BN is the batch normalization, and Φ is the activation function. In subsequent backbone stages, downsampling and channel expansion are realized by

$$\boldsymbol{F}_{i+1} = \text{Conv}_{3\times 3,s=2}\left(F_i\right), \quad i = 1, 2. \tag{8}$$

Through this dual-branch and gated fusion design, the SSFB adaptively balances spatial details and spectral–spatial coupling, yielding robust, edge-preserving, and discriminative features as the foundation for deep spectral–spatial modeling.

C. Spectral–Spatial SSM

The S4M framework receives the shallow spectral–spatial features from the SSFB and performs advanced context modeling across both spectral and spatial domains. As illustrated in Fig. 2(b), it consists of three primary modules: SMSS, SASS, and CASFA. To further improve spectral–spatial representation, SMSS and SASS adopt a global–local–identity decomposition strategy. For HSI classification, such decomposition is beneficial because accurate recognition depends on both long-range spectral–spatial correlations and subtle local differences.

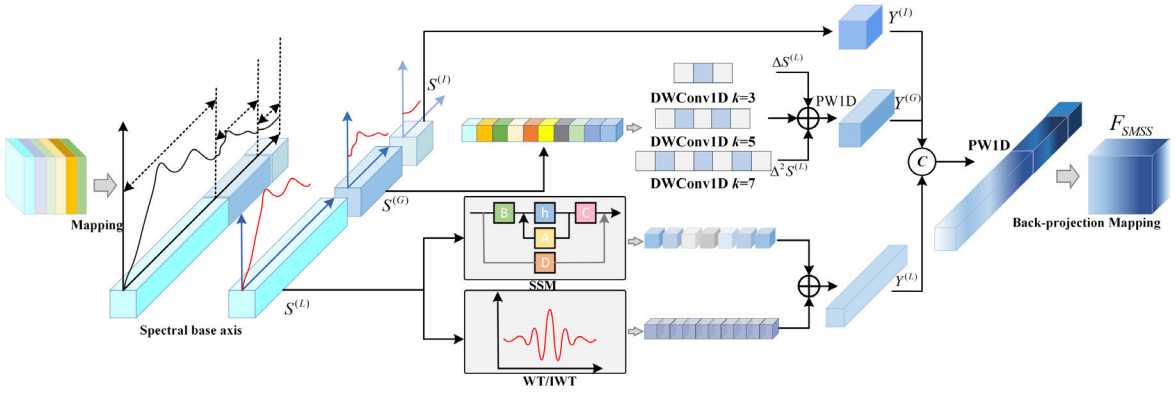


Fig. 4. Architecture of the SMSS module. The input features are projected onto a spectral basis and partitioned into global, local, and identity branches, each undergoing hierarchical spectral modeling and fusion to capture both long-range and localized spectral dependencies.

1) Spectral-Based Multiscale State Space: As shown in Fig. 4, given input features $F \in \mathbb{R}^{B\times C\times H\times W}$, SMSS first projects F along the channel axis onto a low-dimensional spectral basis as follows:

$$\boldsymbol{S} = F W_S, \quad S \in \mathbb{R}^{B\times C^*\times H\times W} \tag{9}$$

where $W_S \in \mathbb{R}^{C\times C^*}$ is a learnable weight matrix. The spectral basis S is partitioned into three segments: global ($S^{(G)}$), local ($S^{(L)}$), and identity ($S^{(I)}$). The partition is performed along the spectral channel ordering in $\boldsymbol{S}$ as three consecutive ranges, rather than a random split. Local band-to-band smoothness and spectral continuity are preserved within each segment. The global segment is processed by a bidirectional SSM and a multiscale wavelet branch as follows:

$$\boldsymbol{Y}^{(G)} = \mathrm{SSM}_{\mathrm{spec}}\left(S^{(G)}\right) + \lambda \cdot \mathrm{IWT}\left(\mathrm{Conv}\left(\mathrm{WT}\left(S^{(G)}\right)\right)\right) \tag{10}$$

where WT(·) and IWT(·) denote the fixed db1/Haar wavelet transform and inverse wavelet transform, Conv represents the learnable wavelet-domain depthwise convolution used for adaptive coefficient refinement, and λ is a learnable scaling factor initialized with a small value to prevent over-injection of wavelet-enhanced responses at the early training stage. The local segment is processed through first- and second-order finite differences, followed by depthwise 1-D convolutions with multiple kernel sizes

$$\boldsymbol{D}^{(1)} = \nabla \boldsymbol{S}^{(L)}, \quad \boldsymbol{D}^{(2)} = \nabla^2 \boldsymbol{S}^{(L)} \tag{11}$$

$$\boldsymbol{Y}^{(L)} = \mathrm{PW1D}\left(\mathrm{DWConv1D}_k\left(\boldsymbol{S}^{(L)}, \boldsymbol{D}^{(1)}, \boldsymbol{D}^{(2)}\right)\right) \tag{12}$$

where ∇ and ∇^2 denote the first- and second-order difference operators, respectively. $\mathrm{DWConv1D}_k$ represents a depthwise 1-D convolution with kernel size k, and PW1D denotes a pointwise 1-D convolution. The identity segment is retained directly as $Y^{(I)} = S^{(I)}$. All segments are concatenated along the channel and fused by pointwise projection, then mapped back to the original feature space and added residually

$$\tilde{\boldsymbol{S}} = \mathrm{PW1D}\left(\mathrm{Concat}\left[\boldsymbol{Y}^{(G)}, \boldsymbol{Y}^{(L)}, \boldsymbol{Y}^{(I)}\right]\right) \tag{13}$$

$$\boldsymbol{F}_{\mathrm{SMSS}} = \boldsymbol{F} + \tilde{\boldsymbol{S}} W_f \tag{14}$$

where W_f is a learnable projection back to the feature axis.

2) Spatial-Axis Multiscale State Space: As shown in Fig. 5, SASS divides $\boldsymbol{F}$ along the channel axis into three parts to enable complementary spatial modeling. This channel partitioning also serves as an implicit redundancy mitigation strategy by enforcing branch specialization and reducing unnecessary interference among heterogeneous operators. The global part $\boldsymbol{F}^{(G)}$ is sequentially modeled along horizontal and vertical spatial axes using state-space modules as follows:

$$\boldsymbol{U}^{(G)} = \mathrm{SSM}_H\left(\boldsymbol{F}^{(G)}\right) + \mathrm{SSM}_W\left(\boldsymbol{F}^{(G)}\right) \tag{15}$$

where SSM_H and SSM_W denote SSMs along height and width dimensions, respectively. In parallel, a 2-D discrete wavelet transform is applied for spatial-frequency enhancement

$$\boldsymbol{V}^{(G)} = \mathrm{IWT}_{2D}\left(\mathrm{Conv}_{\mathrm{sub}}\left(\mathrm{WT}_{2D}\left(\boldsymbol{F}^{(G)}\right)\right)\right) \tag{16}$$

where $\mathrm{IWT}_{2D}(\cdot)$ and $\mathrm{WT}_{2D}(\cdot)$ are fixed db1/Haar-based 2-D wavelet transform and inverse transform, $\mathrm{Conv}_{\mathrm{sub}}$ denotes the learnable wavelet-domain depthwise subband convolution.

To avoid over-injecting potentially redundant wavelet responses, we introduce a learnable scaling factor γ to control the injection strength. The aggregated global output is as $\boldsymbol{Y}^{(G)} = \boldsymbol{U}^{(G)} + \gamma \cdot \boldsymbol{V}^{(G)}$, where γ is initialized with a small value (0.1) and optimized end-to-end, so that the model can progressively increase the wavelet contribution only when it is beneficial. The local part is processed by parallel depthwise separable convolutions with multiple kernel sizes

$$\boldsymbol{Y}^{(L)} = \mathrm{PW}_{1\times 1}\left(\mathrm{Concat}_j\left[\mathrm{DWConv}_{k_j\times k_j}\left(\boldsymbol{F}_j^{(L)}\right)\right]\right). \tag{17}$$

Depthwise separable convolutions emphasize short-range locality while constraining cross-channel mixing, which further helps avoid over-parameterized duplication in local modeling.

The identity part is preserved as $\boldsymbol{Y}^{(I)} = \boldsymbol{F}^{(I)}$ to maintain stable information flow. The outputs are concatenated and fused

$$\tilde{\boldsymbol{F}} = \mathrm{PW}_{1\times 1}\left(\mathrm{Concat}\left[\boldsymbol{Y}^{(G)}, \boldsymbol{Y}^{(L)}, \boldsymbol{Y}^{(I)}\right]\right). \tag{18}$$

The projection acts as a lightweight channel reorganization module that adaptively remixes and compresses branch outputs, thereby suppressing redundant channels and preserving

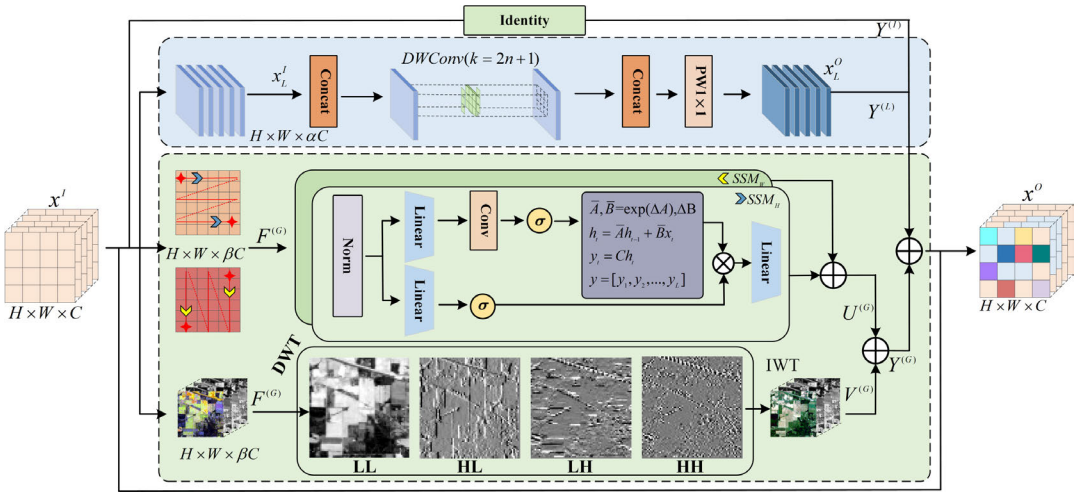


Fig. 5. Architecture of the SASS module. The input features are split into global, local, and identity partitions, which are processed by sequential state-space modeling, multiscale wavelet analysis, and parallel convolutions to extract comprehensive spatial context.

complementary cues with minimal overhead. Finally, a residual connection is applied to ensure fidelity

$$\boldsymbol{F}_{\text{SASS}} = \boldsymbol{F} + \tilde{\boldsymbol{F}}. \tag{19}$$

3) Cross-Axis Spectral–Spatial Feature Alignment: As shown in Fig. 2(e), after axis-wise decomposition, spectral and spatial branches may produce features with different channel responses and semantic emphases. Direct addition or concatenation may therefore introduce feature mismatch. To address this issue, CASFA aligns the decomposed spectral and spatial representations before fusion. Through bidirectional attention and adaptive channel-wise gating, spectral features can guide spatial representations, while spatial features can refine spectral representations. This makes the fusion process more targeted than simple summation, concatenation, or parallel fusion. We first apply global average pooling (GAP) to obtain channel descriptors $\mathbf{x}_s = \text{GAP}(\boldsymbol{F}_{\text{SMSS}})$ and $\mathbf{x}_p = \text{GAP}(\boldsymbol{F}_{\text{SASS}})$. These descriptors are then projected to queries, keys, and values for bidirectional cross-attention. In implementation, the projections are realized by learnable pointwise convolutions, which are equivalent to linear maps on channel descriptors. For the two directions $s \leftarrow p$ and $p \leftarrow s$, we use independent projection parameters to allow asymmetric alignment.

Specifically, these are projected to queries, keys, and values for cross-attention as follows:

$$q_s = \mathbf{x}_s\mathbf{W}_Q^s, \quad k_p = \mathbf{x}_p\mathbf{W}_K^p, \quad v_p = \mathbf{x}_p\mathbf{W}_V^p \tag{20}$$

$$q_p = \mathbf{x}_p\mathbf{W}_Q^p, \quad k_s = \mathbf{x}_s\mathbf{W}_K^s, \quad v_s = \mathbf{x}_s\mathbf{W}_V^s \tag{21}$$

where $\{\mathbf{W}_Q^{(\cdot)}, \mathbf{W}_K^{(\cdot)}, \mathbf{W}_V^{(\cdot)}\}$ denote learnable pointwise projection weights. All projection weights are randomly initialized using standard convolution-layer initialization and are updated end-to-end via back-propagation during training together with the whole network.

The bidirectional attention is calculated as

$$\mathbf{a}_{s\leftarrow p} = \text{softmax}\left(\frac{q_s k_p^\top}{\sqrt{d}}\right), \quad \hat{\mathbf{x}}_s = \mathbf{a}_{s\leftarrow p} v_p \tag{22}$$

$$\mathbf{a}_{p\leftarrow s} = \text{softmax}\left(\frac{q_p k_s^\top}{\sqrt{d}}\right), \quad \hat{\mathbf{x}}_p = \mathbf{a}_{p\leftarrow s} v_s. \tag{23}$$

The attention-enhanced descriptors are further converted to channel-wise gates for dynamic fusion

$$\gamma_s = \sigma\left(\text{MLP}_s\left(\hat{\mathbf{x}}_s\right)\right), \quad \gamma_p = \sigma\left(\text{MLP}_p\left(\hat{\mathbf{x}}_p\right)\right) \tag{24}$$

$$\tilde{\boldsymbol{F}}_s = \boldsymbol{F}_{\text{SMSS}} \odot \gamma_p^{\uparrow}, \quad \tilde{\boldsymbol{F}}_p = \boldsymbol{F}_{\text{SASS}} \odot \gamma_s^{\uparrow} \tag{25}$$

where $\odot$ denotes element-wise multiplication and $\uparrow$ indicates broadcasting to match feature dimensions. The gating networks $\text{MLP}_s(\cdot)$ and $\text{MLP}_p(\cdot)$ are learnable modules and are optimized jointly with the queries, keys, and values projections. A trainable coefficient $\alpha = \sigma(\text{MLP}_\alpha([\mathbf{x}_s, \mathbf{x}_p]))$ is learned for convex fusion

$$\boldsymbol{F}_{\text{CASFA}} = \boldsymbol{F}_{in} + \text{PW}_{1\times1}\left(\alpha^{\uparrow} \odot \tilde{\boldsymbol{F}}_s + (1-\alpha)^{\uparrow} \odot \tilde{\boldsymbol{F}}_p\right) \tag{26}$$

where $\boldsymbol{F}_{in}$ represents the residual shortcut.

D. Lightweight Feature Pyramid Decoder

Since features from different stages retain complementary properties, such as fine-grained local details at lower levels and rich semantics at higher levels, multiscale context aggregation is necessary for accurate dense HSI classification. Meanwhile, objects with diverse scales and shapes require the decoder to recover high-resolution prediction maps while avoiding excessive computational overhead. To address these challenges, we propose an LFPD for efficient multiscale fusion and progressive reconstruction. In this work, lightweight refers specifically to the decoder-level lightweightness, i.e., reducing the additional parameters and FLOPs introduced by the decoding stage. This is achieved through 1×1 lateral channel alignment, top-down additive fusion, bilinear upsampling, depthwise separable smoothing convolutions, and ASPP-lite-based multiscale aggregation.

Specifically, as shown in Fig. 2, the decoder receives multistage features $\{f_1, f_2, f_3\}$ and employs a top-down feature pyramid network [49] with lateral connections to integrate cross-scale information while preserving high-resolution details. To further enhance multiscale semantic aggregation with minimal computational overhead, an ASPP-lite [50] is

applied to the highest-resolution feature map, followed by a 1×1 prediction head to produce dense pixel-wise logits. Let $f_1 \in \mathbb{R}^{B\times C_1\times H/16\times W/16}$, $f_2 \in \mathbb{R}^{B\times C_2\times H/32\times W/32}$, and $f_3 \in \mathbb{R}^{B\times C_3\times H/64\times W/64}$ denote the features from three consecutive encoder stages. Each is first projected to a common decoder channel dimension D via 1×1 convolutions, yielding p_1^0, p_2^0, and p_3^0 as follows:

$$p_3^0 = \phi_{1\times1}^{(3)}(f_3), \quad p_2^0 = \phi_{1\times1}^{(2)}(f_2), \quad p_1^0 = \phi_{1\times1}^{(1)}(f_1) \tag{27}$$

where $\phi_{1\times1}^{(i)}$ denotes a 1×1 convolution for lateral channel alignment. A top-down pathway then progressively upsamples higher-level features and merges them with the adjacent lateral features via element-wise addition

$$p_2 = p_2^0 + \mathrm{Up}\left(p_3^0; h_2, w_2\right), \quad p_1 = p_1^0 + \mathrm{Up}\left(p_2; h_1, w_1\right) \tag{28}$$

where $\mathrm{Up}(\cdot; h, w)$ indicates bilinear upsampling to spatial size (h, w). To mitigate aliasing artifacts from upsampling and promote local consistency, depthwise separable 3×3 convolutions are applied

$$\hat{p}_2 = \psi_{\mathrm{DW}-3\times3}(p_2), \quad \hat{p}_1 = \psi_{\mathrm{DW}-3\times3}(p_1) \tag{29}$$

where $\psi_{\mathrm{DW}-3\times3}$ denotes a depthwise separable convolution. A standard 3×3 convolution is then applied to $\hat{p}_1$ to yield the fused feature map $y_{\mathrm{FPN}} = \phi_{3\times3}(\hat{p}_1)$. To efficiently aggregate multiscale context, an ASPP-lite module is applied to y_{FPN}, consisting of a 1×1 branch and several parallel depthwise separable atrous convolutions with dilation rates $r \in \mathcal{R}$. ASPP-lite takes multiscale features from the three stages as input. For each stage feature branch, we apply depthwise-separable atrous convolutions with dilation rates $r = \{1, 2, 3\}$, together with a pointwise projection for lightweight channel mixing

$$g_0 = \phi_{1\times1}(y_{\mathrm{FPN}}), \quad g_r = \phi_{\mathrm{DW}-3\times3}^{(d=r)}(y_{\mathrm{FPN}}) \tag{30}$$

$$y_{\mathrm{LFPD}} = \phi_{1\times1}\left(\mathrm{Concat}\left[g_0, \{g_r\}_{r\in\mathcal{R}}\right]\right) \tag{31}$$

where $\phi_{\mathrm{DW}-3\times3}^{(d=r)}$ denotes a depthwise separable 3×3 convolution with dilation r, and Concat$[\cdot]$ concatenates along the channel dimension. Finally, a 1×1 prediction head generates the final pixel-wise class logits

$$Z = \phi_{1\times1}^{\mathrm{cls}}(y_{\mathrm{LFPD}}), \quad Z \in \mathbb{R}^{B\times C_{\mathrm{cls}}\times H\times W} \tag{32}$$

where $\phi_{1\times1}^{\mathrm{cls}}$ is the classification head and C_{cls} is the number of object classes. The decoder propagates cross-scale information and achieves spatial alignment, transforming mid-level spectral–spatial features into high-resolution.

E. Loss Function

Pixel-wise supervised learning is employed to optimize the network parameters. To promote stable convergence and address class imbalance as well as hard-to-classify pixels, a staged loss strategy is adopted. Specifically, the network is initially trained with the standard cross-entropy (CE) Loss for several epochs as a warm-up phase, after which training proceeds with the focal loss to further enhance robustness.

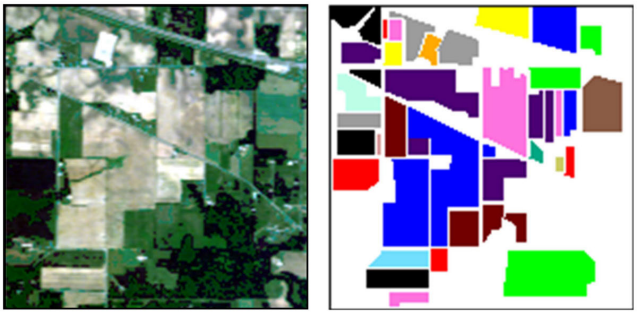
Let the predicted logits be $Z \in \mathbb{R}^{B\times C_{\mathrm{cls}}\times H\times W}$, with corresponding softmax probabilities $P = \mathrm{softmax}(Z)$, and ground-truth pixel labels $y \in \{1, \ldots, C_{\mathrm{cls}}\}$. Denote Ω as the set of valid pixels over the batch and spatial dimensions. The warm-up loss is defined as

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{|\Omega|}\sum_{(b,i,j)\in\Omega} \omega_{y_{(b,i,j)}} \log\left(P_{b,y_{(b,i,j)},i,j} + \varepsilon\right) \tag{33}$$

where $\omega_{y_{(b,i,j)}}$ is an optional class weight to address class imbalance, b, i, and j index the batch, height, and width dimensions, respectively, and ε is a small constant to prevent numerical instability. After the warm-up, focal loss is used to emphasize hard examples and mitigate class imbalance

$$\mathcal{L}_{\mathrm{Focal}} = -\frac{1}{|\Omega|}\sum_{(b,i,j)\in\Omega} \alpha_t (1 - p_t)^{\gamma} \log(p_t + \varepsilon) \tag{34}$$

where $p_t = P_{b,y_{(b,i,j)},i,j}$ denotes the predicted probability of the true class at pixel (i, j) of sample b, $\alpha_t \in [0, 1]$ is a class-balancing factor, and $\gamma > 0$ is the focusing parameter. The $(1 - p_t)^{\gamma}$ acts as a modulating factor to focus training on hard, misclassified pixels. The logarithm is stabilized by adding a small ε to avoid evaluating log(0).



(a) (b)

Fig. 6. IP dataset. (a) False-color image. (b) Ground truth.

IV. Experiments and Analysis

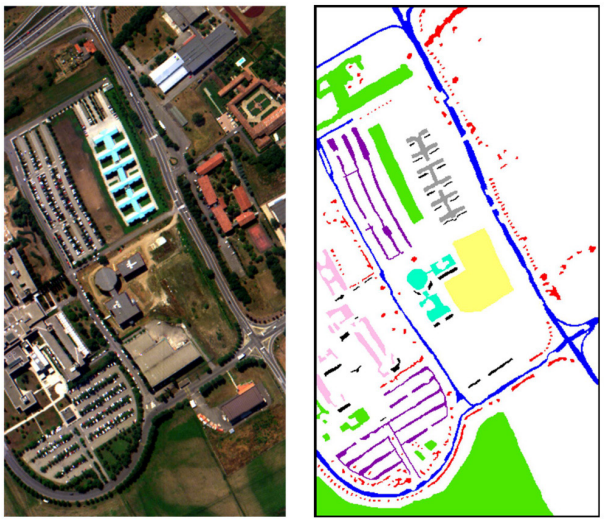
A. Dataset Description

To comprehensively evaluate the classification performance of the proposed WD4SNet, we conduct experiments on four widely used HSI datasets: Indian Pines (IP) [12], Pavia University (PU) [12], Salinas (SA) [51], and University of Houston (HU) [51]. All experiments strictly follow the official training and testing splits from the original references to ensure fair comparison.

1) Indian Pines [12]: The IP dataset is a classic benchmark for HSI classification, collected by the AVIRIS sensor over northwestern Indiana, USA. It consists of 224 spectral bands spanning the 0.4–2.5-μm range, with a spatial dimension of 145×145 pixels and a ground sampling distance of 20 m. The scene primarily contains agricultural fields and woodlands, featuring crops such as corn, soybeans, and wheat at various growth stages. High spectral similarity among certain classes poses considerable challenges for accurate classification. Fig. 6 visualizes the data and label distribution; corresponding statistics are listed in Table I.

TABLE I
CLASS DISTRIBUTION OF THE IP DATASET

ID	Color	Class	Total	Train	Test
1		Alfalfa	54	20	34
2		Corn-notill	1434	20	1414
3		Corn-mintill	834	20	814
4		Corn	234	20	214
5		Grass-pasture	497	20	477
6		Grass-trees	747	20	727
7		Grass-pasture-moved	26	10	16
8		Hay-windrowed	489	20	469
9		Oats	20	5	15
10		Soybeans-notill	968	20	948
11		Soybeans-mintill	2468	20	2448
12		Soybeans-clean	614	20	594
13		Wheat	212	20	192
14		Woods	1294	20	1274
15		Bldg-grass-tree-drives	380	20	360
16		Stone-steel-towers	95	20	75
Total			**10366**	**295**	**10071**



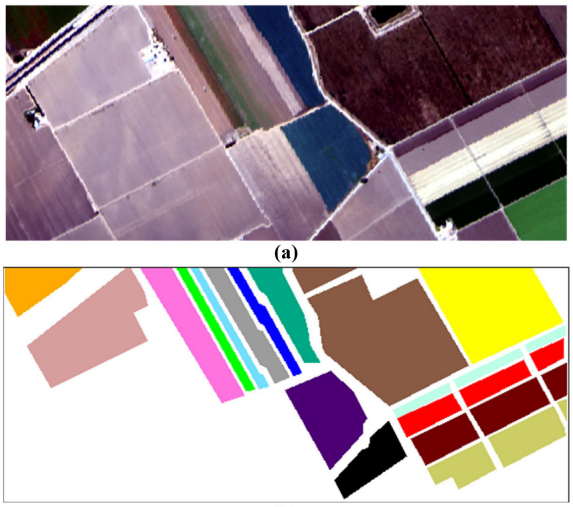
(a) (b)

Fig. 7. PU dataset. (a) False-color image. (b) Ground truth.

TABLE II
CLASS DISTRIBUTION OF THE PU DATASET

ID	Color	Class	Total	Train	Test
1		Asphalt	6631	20	6611
2		Meadows	18649	20	18629
3		Gravel	2099	20	2079
4		Trees	3064	20	3044
5		Painted metal sheets	1345	20	1325
6		Bare soil	5029	20	5009
7		Bitumen	1330	20	1310
8		Self-blocking bricks	3682	20	3662
9		Shadows	947	20	927
Total			**42776**	**180**	**42596**

2) Pavia University [12]: Acquired by the ROSIS sensor over Pavia, Italy, this dataset comprises 115 spectral bands (after noisy bands removal) covering the 0.43–0.86-μm range. The image has a spatial size of 610 $\times$ 340 pixels with a fine spatial resolution of 1.3 m. The urban scene encompasses man-made structures (buildings, asphalt, and roads) and natural elements (trees, bare soil, and shadows), making it a representative benchmark for urban land-cover classification. An overview of this dataset is given in Fig. 7 and Table II.



(a) (b)

Fig. 8. SA dataset. (a) False-color image. (b) Ground truth.

TABLE III
CLASS DISTRIBUTION OF THE SA DATASET

ID	Color	Class	Total	Train	Test
1		Brocoli green weeds 1	2009	20	1989
2		Brocoli green weeds 2	3726	20	3706
3		Fallow	1976	20	1956
4		Fallow rough plow	1394	20	1374
5		Fallow smooth	2678	20	2658
6		Stubble	3959	20	3939
7		Celery	3579	20	3559
8		Grapes untrained	11271	20	11251
9		Soil vineyard develop	6203	20	6183
10		Corn senesced green weeds	3278	20	3258
11		Lettuce romaine 4wk	1068	20	1048
12		Lettuce romaine 5wk	1927	20	1907
13		Lettuce romaine 6wk	916	20	896
14		Lettuce romaine 7wk	1070	20	1050
15		Vinyard untrained	7268	20	7248
16		Vinyard vertical trellis	1807	20	1787
Total			**54129**	**320**	**53809**

3) Salinas [51]: The SA dataset was collected by AVIRIS over the SA Valley in California, USA. It contains 224 spectral bands and has a spatial size of 512 $\times$ 217 pixels at a resolution of 3.7 m. The scene is dominated by agricultural fields with various vegetable crops (e.g., broccoli, lettuce, and celery) arranged in regular patterns. The dataset features low interclass variance and minimal noise, providing a clean benchmark for assessing discriminative power. An overview of this dataset is given in Fig. 8 and Table III.

4) University of Houston [51]: The HU dataset originates from the 2013 IEEE GRSS Data Fusion Contest, captured by the ITRES CASI-1500 sensor over the HU campus and adjacent urban areas. It consists of 144 spectral bands spanning 0.38–1.05 μm, with a spatial size of 349 $\times$ 1905 pixels and a resolution of 2.5 m. The complex scene includes diverse land-cover types such as healthy grass, stressed grass, trees, buildings, and roads. Its large spatial extent and detailed ground truth annotations make it particularly suitable for evaluating classification methods in urban and semi-urban environments. An overview of this dataset is given in Fig. 9 and Table IV.

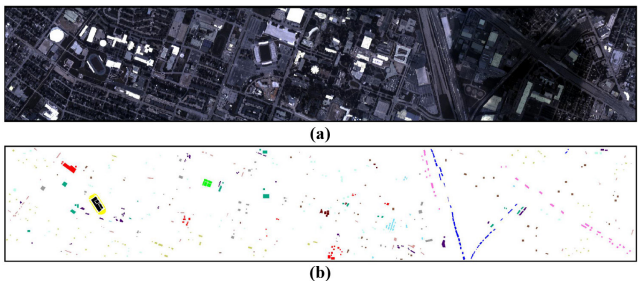


Fig. 9. HU dataset. (a) False-color image. (b) Ground truth.

TABLE IV
CLASS DISTRIBUTION OF THE HU DATASET

ID	Color	Class	Total	Train	Test
1		Healthy grass	1251	20	1231
2		Stressed grass	1254	20	1234
3		Synthetic grass	697	20	677
4		Trees	1244	20	1224
5		Soil	1242	20	1222
6		Water	325	20	305
7		Residential	1268	20	1248
8		Commercial	1244	20	1224
9		Road	1252	20	1232
10		Highway	1227	20	1207
11		Railway	1235	20	1215
12		Parking Lot1	1233	20	1213
13		Parking Lot2	469	20	449
14		Tennis court	428	20	408
15		Running track	660	20	640
Total			**15029**	**300**	**14729**

B. Evaluation Metrics

To quantitatively assess the performance of the proposed classification method, three widely recognized metrics are utilized: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. These metrics comprehensively evaluate both global and per-class classification effectiveness. 1) OA is defined as the proportion of correctly classified pixels among all test pixels

$$\mathrm{OA} = \frac{\sum_{c=1}^{C} n_{cc}}{N} \tag{35}$$

where C is the number of classes, n_{cc} denotes the number of pixels correctly classified as class c, and N is the total number of test pixels. 2) AA is calculated as the arithmetic mean of per-class accuracies

$$\mathrm{AA} = \frac{1}{C}\sum_{c=1}^{C}\frac{n_{cc}}{n_c} \tag{36}$$

where n_c is the total number of ground-truth pixels belonging to class c. 3) Kappa coefficient (κ) measures the agreement between predictions and ground truth, correcting for chance agreement. It is formulated as

$$\kappa = \frac{\mathrm{OA} - \mathrm{P}_e}{1 - \mathrm{P}_e} \tag{37}$$

where P_e is the expected agreement by chance, given by

$$\mathrm{P}_e = \frac{\sum_{c=1}^{C} n_c^{\mathrm{pred}} n_c}{N^2} \tag{38}$$

with n_c^{pred} is the number of pixels predicted as class c. Higher κ values indicate stronger consistency with the ground truth.

C. Implementation Details

All experiments are conducted on a workstation equipped with an NVIDIA RTX A5000 GPU (24 GB memory) and an Intel Xeon Gold 6330 CPU. The proposed framework is implemented in PyTorch. During training, the AdamW optimizer is used for 150 epochs with a mini-batch size of 64. The initial learning rate is set to 0.001 and decayed by a factor of 0.1 at 60% and 90% of the total epochs. A weight decay of 0.0001 is applied to all trainable parameters to mitigate overfitting. All inputs are normalized to zero mean and unit variance on a per-band basis before training. Random horizontal and vertical flips are used for data augmentation during training to enhance model generalization. In the Mamba module, the state dimension of the SS2D operator is set to 16, and the expansion ratio was set to 2, consistent with the configuration of EfficientvMamba [52]. All other hyperparameters are selected based on grid search and recommendations from related literature to ensure fair and reproducible comparison.

D. Baseline Methods for Comparison

To comprehensively evaluate the effectiveness of WD4SNet, we compare it with a diverse set of baseline methods covering four representative categories: traditional machine learning approaches, CNNs-based architectures, Transformer-based models, and recent Mamba-based techniques.

1) *SVM [9]:* Serves as a classical SVM classifier, performing nonlinear classification using kernel functions on raw hyperspectral data.
2) *ASSMN [28]:* Fuses spectral features with LSTM and spatial features with ConvLSTM for end-to-end adaptive spectral–spatial learning.
3) *3D-CNN [23]:* Utilizes 3-D convolutional neural networks to jointly extract and fuse spectral and spatial features from hyperspectral cubes.
4) *A2S2K [29]:* Leverages attention-driven selective 3-D convolutions and spectral attention within residual blocks for adaptive spectral–spatial representation.
5) *SSFTT [31]:* Integrates convolutional tokenization and a Transformer encoder to learn high-level semantic representations for classification.
6) *HiT [32]:* Embeds convolutional projections and a Conv-Permutator module within a Transformer to capture fine-grained spectral and spatial information.
7) *SPRLT [33]:* Employs a local Transformer with spatial patch recovery to enable adaptive spectral–spatial representation and global context modeling.
8) *MASSF [34]:* Introduces memory tokens and a memory-augmented Transformer encoder to enhance feature mixing and deep spectral–spatial representation.
9) *MIM [44]:* Utilizes Mamba-cross-scan and multiscale semantic token fusion within a Mamba-based architecture for efficient feature extraction.

TABLE V

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, KAPPA, AND THE ACCURACY OF EACH CLASS FOR THE IP. THE BEST ACCURACIES ARE PRESENTED BLOD

Class	SVM	ASSMN	3D-CNN	A2S2K	SSFTT	HiT	SPRLT	MASSF	MIM	DBMG	FAHM	WD4SNet
1	98.54	68.50	**100.00**	98.55	87.86	**100.00**	95.65	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
2	58.25	73.62	55.64	76.22	87.74	76.63	83.51	86.50	**97.96**	93.27	95.46	95.93
3	86.34	74.88	70.49	96.17	92.69	98.27	98.63	92.82	99.84	99.12	**99.85**	99.05
4	92.07	94.27	97.36	93.55	**100.00**	97.70	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
5	81.18	89.01	80.55	85.96	94.92	95.90	96.87	87.30	91.52	99.10	94.29	**100.00**
6	72.78	80.69	93.75	85.49	84.43	98.45	**99.68**	95.15	97.36	96.55	96.90	93.96
7	99.91	**100.00**	**100.00**	96.92	**100.00**	**100.00**	**100.00**	95.62	**100.00**	**100.00**	**100.00**	**100.00**
8	98.99	**100.00**	93.30	96.80	91.67	**100.00**	**100.00**	**100.00**	**100.00**	99.99	**100.00**	**100.00**
9	**100.00**	**100.00**	98.75	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
10	81.91	85.34	84.30	90.34	72.45	83.30	84.63	**98.12**	91.58	97.45	96.23	95.85
11	55.42	76.89	74.99	69.23	90.94	71.79	94.23	89.73	88.78	94.40	96.70	**96.98**
12	65.52	71.87	64.84	67.54	90.24	81.85	92.49	94.84	95.67	89.84	93.68	**95.67**
13	**100.00**	88.08	98.97	**100.00**	**100.00**	97.84	98.10	98.71	93.20	98.06	98.06	97.09
14	87.97	71.24	94.42	99.92	**100.00**	98.63	**100.00**	98.77	99.91	99.19	99.73	99.81
15	98.67	96.01	89.15	99.73	99.40	99.73	**100.00**	99.70	98.96	**100.00**	98.73	**100.00**
16	**100.00**	98.80	**100.00**	98.63	**100.00**	98.63	87.23	70.21	89.36	**100.00**	93.62	**100.00**
OA(%)	73.95	79.38	76.33	83.17	91.22	86.42	93.76	92.36	97.23	97.07	97.16	**97.36**
AA(%)	83.54	88.63	88.17	88.48	88.82	88.55	93.62	90.07	95.21	95.48	96.95	**98.40**
Kappa×100	71.04	76.75	73.47	81.06	89.93	84.68	92.78	91.27	93.92	95.58	96.66	**96.90**

TABLE VI

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, KAPPA, AND THE ACCURACY OF EACH CLASS FOR THE PU. THE BEST ACCURACIES ARE PRESENTED IN BLOD

Class	SVM	ASSMN	3D-CNN	A2S2K	SSFTT	HiT	SPRLT	MASSF	MIM	DBMG	FAHM	WD4SNet
1	91.23	79.43	77.85	87.27	81.54	83.79	90.20	85.96	98.53	97.19	96.43	**98.69**
2	69.52	92.83	95.81	95.56	96.55	96.58	**99.21**	94.52	98.27	97.40	98.26	99.11
3	72.65	98.73	91.31	95.90	94.05	94.68	96.70	96.19	99.00	96.95	**99.13**	97.89
4	**100.00**	70.87	70.54	65.69	68.71	77.80	82.22	73.26	89.37	90.99	92.96	93.99
5	86.12	95.14	94.36	95.98	98.46	92.36	98.71	97.53	**99.63**	92.05	98.08	89.76
6	99.19	**100.00**	99.64	**100.00**	99.96	99.14	99.13	99.70	99.98	**100.00**	99.52	99.85
7	97.16	96.33	98.67	99.06	99.45	**100.00**	99.84	97.97	99.81	**100.00**	98.89	99.03
8	78.88	86.65	98.98	88.74	93.64	98.35	93.94	92.21	98.92	99.28	98.05	**99.53**
9	98.19	76.65	78.37	78.15	79.71	72.02	82.64	75.14	78.62	85.41	91.25	**96.14**
OA(%)	78.51	89.61	91.40	92.02	92.05	93.07	95.69	91.95	97.73	97.07	97.66	**98.47**
AA(%)	88.32	88.63	88.17	88.48	88.82	88.55	93.62	90.07	95.21	95.48	96.95	**97.11**
Kappa×100	75.57	86.38	88.66	89.46	89.50	90.86	94.25	89.41	96.94	96.07	96.85	**97.92**

10) *DBMG [45]:* Combines a Mamba branch for long-range dependencies with a GCN branch that captures non-Euclidean spatial relationships.
11) *FAHM [46]:* Employs hierarchical frequency-aware modules and bidirectional adaptive Mamba blocks for robust and generalizable spectral–spatial modeling.

E. Comparison Results and Analysis

1) Quantitative Analysis: To objectively assess the classification performance, we report the OA, AA, and Kappa coefficient for all competing methods on the four widely used hyperspectral benchmarks: IP, PU, SA, and HU. All results are averaged over three independent runs to mitigate random fluctuations, and the same training/testing splits are adopted for all methods to ensure a fair comparison. The detailed class-wise and overall results are summarized in Tables V–VIII. On the IP dataset (see Table V), WD4SNet achieves the best results, with OA of 97.36%, AA of 98.40%, and Kappa coefficient of 96.90. Compared to the strongest baseline FAHM [46] (OA: 97.16%, Kappa: 96.66), WD4SNet improves OA and Kappa by 0.2% and 0.24, respectively. For AA, WD4SNet surpasses DBMG [45] (95.48%) by 2.92%. Importantly, WD4SNet attains the highest class-wise accuracy in 12 out of 16 categories, demonstrating not only superior global performance but also strong per-class consistency, especially for challenging classes with high spectral similarity. On the PU dataset (see Table VI), WD4SNet also yields the highest OA (98.47%), AA (97.11%), and Kappa (97.92). The OA and Kappa improvements over the best baseline MIM [41] (OA: 97.73%, Kappa: 96.94) are 0.74% and 0.98, respectively. Compared with FAHM [46], WD4SNet achieves a slightly higher AA by 0.16%. The model maintains high class-wise accuracy even on classes with limited sample sizes, reflecting its robustness against data imbalance. For the SA dataset (see Table VII), WD4SNet achieves an OA of 98.77%, AA of 98.78%, and Kappa of 98.63, outperforming FAHM [46] by 0.83% (OA), 0.79% (AA), and 0.93 (Kappa). The model secures the best class-wise accuracy in 11 out of 16 categories, further confirming its stable and reliable performance across diverse land cover types. On the HU dataset (see Table VIII), WD4SNet obtains OA of 96.40%, AA of 96.55%, and Kappa of 96.09, exceeding FAHM [46] by 1.23% (OA), 0.44% (AA), and 1.32 (Kappa). The performance advantage is particularly notable on minority or spectrally confusing classes, which are traditionally difficult for most baselines. The HU dataset

TABLE VII

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, KAPPA, AND THE ACCURACY OF EACH CLASS FOR THE SA. THE BEST ACCURACIES ARE PRESENTED IN BLOD

Class	SVM	ASSMN	3D-CNN	A2S2K	SSFTT	HiT	SPRLT	MASSF	MIM	DBMG	FAHM	WD4SNet
1	97.55	98.70	**99.95**	98.15	99.60	98.95	99.25	99.85	99.60	99.85	99.85	99.75
2	96.43	95.78	99.95	99.97	**100.00**	99.97	98.60	98.89	**100.00**	96.22	97.54	99.97
3	90.26	99.54	98.98	97.61	99.49	99.95	92.42	**100.00**	99.49	**100.00**	**100.00**	**100.00**
4	88.48	90.68	82.51	97.25	97.74	95.30	94.22	**98.91**	97.74	94.03	98.84	97.26
5	88.85	94.57	86.73	89.13	93.98	83.51	91.53	93.00	93.98	92.40	92.55	**96.92**
6	99.47	**100.00**	95.47	92.96	92.21	96.61	99.80	99.72	92.21	98.15	99.31	98.24
7	97.34	94.34	99.94	92.21	**100.00**	98.63	99.33	97.64	**100.00**	90.64	97.25	**100.00**
8	86.45	87.65	85.84	89.16	89.03	85.26	80.52	92.97	89.03	93.46	97.40	**99.85**
9	97.24	93.19	98.27	98.32	**100.00**	**100.00**	98.51	97.04	**100.00**	**100.00**	**100.00**	99.81
10	86.43	82.80	94.37	88.22	97.91	97.28	96.18	97.33	97.91	85.67	97.45	**98.14**
11	99.43	93.67	99.43	91.30	**100.00**	99.53	98.39	99.81	**100.00**	99.81	**100.00**	**100.00**
12	98.80	80.86	86.91	87.58	98.37	92.28	**99.17**	89.20	98.37	98.48	95.33	96.87
13	61.91	95.47	99.34	94.59	98.77	98.68	95.58	97.21	98.77	95.65	96.43	**99.43**
14	74.74	97.26	94.43	77.92	93.43	94.25	96.51	92.38	93.43	**98.67**	98.38	98.24
15	81.50	92.70	93.29	99.35	99.10	88.08	98.64	93.79	99.10	**99.81**	97.74	95.93
16	99.11	**100.00**	**100.00**	**100.00**	**100.00**	99.94	99.61	99.61	**100.00**	**100.00**	**100.00**	**100.00**
OA(%)	90.57	92.45	93.52	93.81	96.29	93.42	94.11	95.97	96.29	96.03	97.94	**98.77**
AA(%)	90.18	92.78	94.23	93.18	97.32	95.73	95.83	96.34	97.32	96.24	97.99	**98.78**
Kappa×100	89.52	91.63	92.82	93.13	95.88	92.70	93.46	95.52	95.88	95.59	97.70	**98.63**

TABLE VIII

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, KAPPA, AND THE ACCURACY OF EACH CLASS FOR THE HU. THE BEST ACCURACY ARE PRESENTED BLOD

Class	SVM	ASSMN	3D-CNN	A2S2K	SSFTT	HiT	SPRLT	MASSF	MIM	DBMG	FAHM	WD4SNet
1	82.78	83.92	90.20	97.18	85.46	84.76	**100.00**	97.18	89.92	82.22	97.95	97.01
2	76.42	55.19	89.16	97.36	87.51	87.21	86.89	97.36	85.52	**99.52**	94.04	84.64
3	97.64	35.45	**100.00**	98.06	99.16	97.99	82.61	98.06	99.84	**100.00**	98.54	97.49
4	48.77	35.21	94.26	85.14	97.43	75.46	94.23	85.14	94.42	**99.95**	93.38	91.73
5	**100.00**	96.97	96.68	**100.00**	**100.00**	99.83	97.56	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
6	98.03	89.18	93.52	99.59	77.69	**100.00**	54.11	99.59	**100.00**	95.74	**100.00**	93.87
7	86.70	79.97	71.12	95.12	68.95	79.15	94.17	95.12	93.10	91.52	89.56	**96.60**
8	70.51	65.28	54.07	89.69	86.54	90.54	89.49	89.69	86.77	92.35	92.87	**96.99**
9	58.93	58.04	69.15	85.58	60.13	81.11	95.12	85.58	91.21	82.96	80.29	**96.64**
10	92.29	89.15	75.91	93.46	88.62	**100.00**	91.03	93.46	**100.00**	79.54	99.91	**100.00**
11	**100.00**	92.59	87.01	**100.00**	89.39	99.32	75.88	**100.00**	97.32	97.55	**100.00**	**100.00**
12	90.60	70.98	69.86	97.40	98.56	93.15	91.40	97.40	99.65	84.52	95.92	**100.00**
13	88.86	79.06	95.03	**100.00**	95.63	91.89	93.29	**100.00**	**100.00**	92.23	**100.00**	**100.00**
14	**100.00**	**100.00**	99.45	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	98.56	**100.00**	**100.00**
15	74.22	73.28	99.82	65.88	97.65	81.64	95.09	99.83	97.76	**100.00**	99.66	93.33
OA(%)	82.30	72.29	82.87	92.82	86.77	89.68	90.33	94.92	94.63	91.91	95.17	**96.40**
AA(%)	83.58	72.93	85.79	93.71	89.47	90.89	96.06	96.03	96.02	93.19	96.11	**96.55**
Kappa×100	80.89	70.09	81.49	91.50	84.89	88.84	89.61	94.50	94.19	91.23	94.77	**96.09**

contains several categories with relatively few test pixels, which usually correspond to small-area targets and are more challenging to classify reliably. To highlight robustness under such minority conditions, we further analyze per-class accuracies for the classes with test pixels < 700 (i.e., Classes 3, 6, 13, 14, and 15). As shown, our method achieves 100% accuracy on five minority classes, including parking lot2 (Class 13, 449) and tennis court (Class 14, 408), and obtains competitive results on synthetic grass, water, and running track, demonstrating strong robustness on small-area categories. Although synthetic grass (Class 3, 677) remains challenging due to its spectral similarity to other grass-related classes, our overall minority-class performance remains competitive. These results indicate that the proposed spectral–spatial global–local modeling and cross-axis alignment help preserve discriminative cues for small targets and reduce confusion under limited-category support. Across all four benchmarks, WD4SNet consistently achieves the highest OA, AA, and Kappa values, validating its effectiveness and generalization ability. Not only does it deliver state-of-the-art average performance, but it also maintains leading class-wise accuracy in the majority of categories, especially on complex or imbalanced datasets. This indicates that WD4SNet can effectively model subtle spectral–spatial differences and is less sensitive to class imbalance or limited training samples. These quantitative results highlight the superiority of WD4SNet in joint spectral–spatial feature extraction, multiscale information integration, and long-range dependency modeling. The significant performance gains, especially on challenging and imbalanced scenarios, demonstrate the practical value and strong robustness of the proposed approach for real-world hyperspectral image classification tasks.

2) Qualitative Analysis: To provide an intuitive understanding of classification performance, we present the true-color images, ground-truth label maps, and visualized classification results for all methods across the four benchmark datasets (see Figs. 10–13). These qualitative comparisons further illustrate the superior capabilities of WD4SNet in spatial context

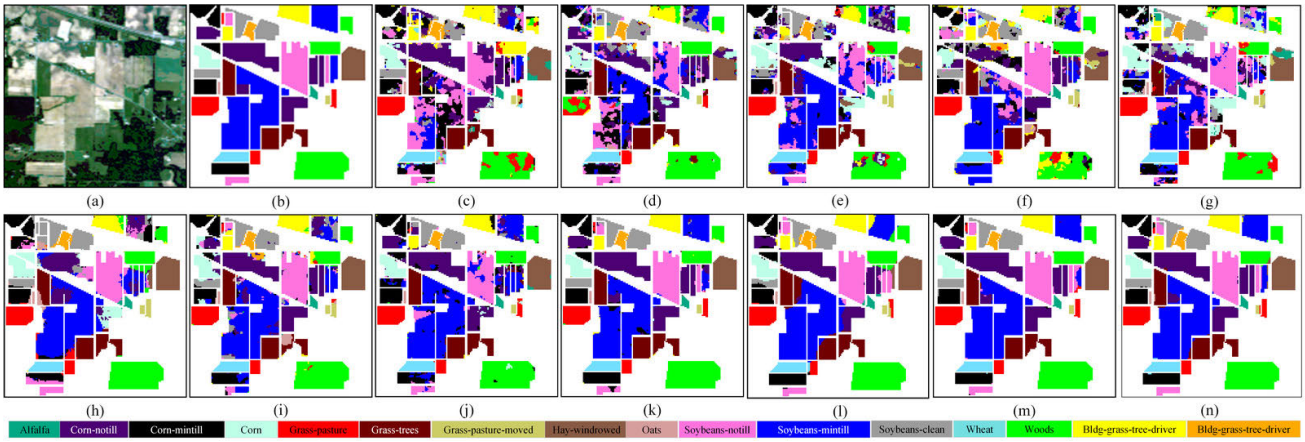


Fig. 10. Classification maps obtained by different methods on IP. (a) False-color image. (b) Ground truth. (c) SVM. (d) ASSMN. (e) 3D-CNN. (f) A2S2K. (g) SSFTT. (h) HiT. (i) SPRLT. (j) MASSF. (k) MIM. (l) DBMG. (m) FAHM. (n) WD4SNet (Ours).

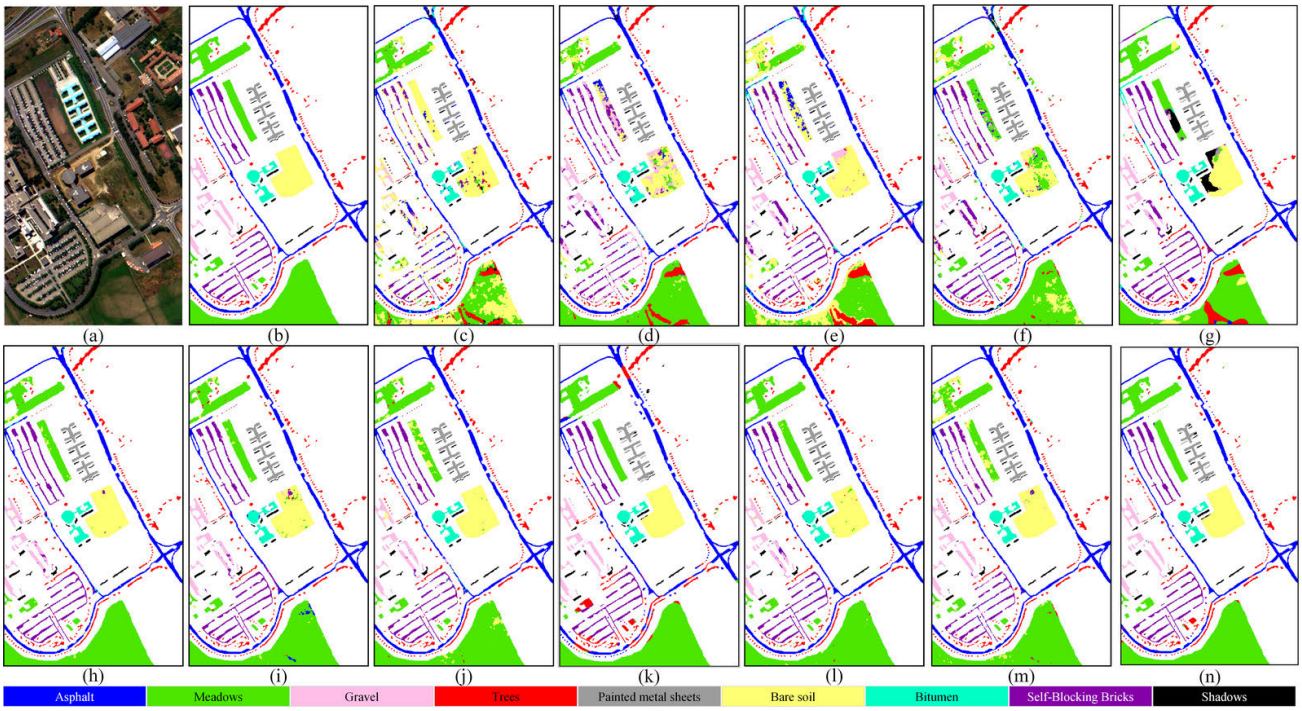


Fig. 11. Classification maps obtained by different methods on PU. (a) False-color image. (b) Ground truth. (c) SVM. (d) ASSMN. (e) 3D-CNN. (f) A2S2K. (g) SSFTT. (h) HiT. (i) SPRLT. (j) MASSF. (k) MIM. (l) DBMG. (m) FAHM. (n) WD4SNet (Ours).

modeling, fine-grained boundary delineation, and robust feature fusion. As shown in Fig. 10, WD4SNet yields the cleanest and most accurate classification map among all approaches. For challenging classes with complex boundaries or small spatial extent (e.g., hay-windrowed and oats), our method produces more continuous and homogeneous regions, with significantly fewer isolated misclassified pixels compared to traditional, CNNs- and Transformer-based methods in Fig. 10(c)–(j). This demonstrates its advanced spatial context modeling and effective multiscale feature fusion. Moreover, WD4SNet substantially reduces confusion between spectrally similar categories such as soybeans-notill versus soybeans-mintill and grass-pasture versus grass-pasture-mowed, highlighting its strength in spectral–spatial feature learning and higher-order dependency capture. For large-area classes (e.g., corn-notill and corn-mintill), the classification map is notably more uniform, with fewer misclassified "spots," reflecting strong global feature integration. Compared to other Mamba-based baselines, WD4SNet better recognizes small and sparse classes (e.g., oats, alfalfa, and grass-pasture-mowed), illustrating superior generalization under limited training data. As depicted in Fig. 11, the classification result of WD4SNet is visually closest to the ground truth, with minimal noise and the most accurate region boundaries. Particularly in shadowed areas, WD4SNet accurately distinguishes shadow pixels, closely matching the ground-truth, which is a common challenge for most methods. For large continuous classes (e.g., asphalt and meadows), WD4SNet yields internally

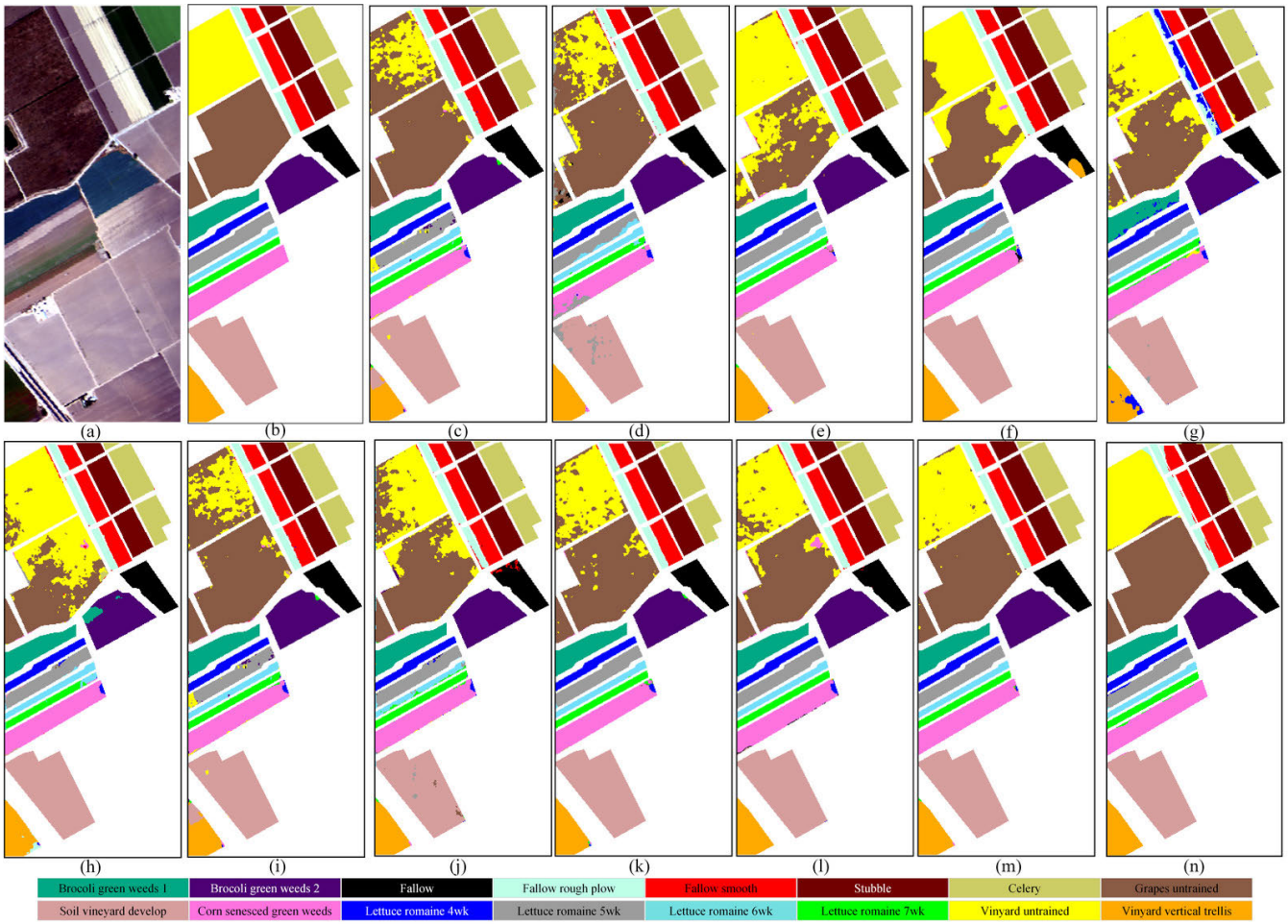


Fig. 12. Classification maps obtained by different methods on SA. (a) False-color image. (b) Ground truth. (c) SVM. (d) ASSMN. (e) 3D-CNN. (f) A2S2K. (g) SSFTT. (h) HiT. (i) SPRLT. (j) MASSF. (k) MIM. (l) DBMG. (m) FAHM. (n) WD4SNet (Ours).

consistent regions without the "salt-and-pepper" noise that persists in other approaches, demonstrating excellent spatial feature aggregation. In small and sparse categories (e.g., painted metal sheets and self-blocking bricks), our method preserves fine details and avoids missing small targets, further highlighting its robustness. While CNNs- and Transformer-based models show some improvements over traditional baselines in large-area classification, they still struggle with small classes and shadow discrimination. In contrast, WD4SNet achieves nearly zero misclassification in large areas and excels in shadowed region classification, underscoring its advanced feature fusion and robustness. As illustrated in Fig. 12, WD4SNet produces sharp and precise boundaries that closely align with the ground truth. Traditional methods yield substantial misclassification, especially in fine-grained classes (e.g., lettuce at different growth stages), resulting in scattered predictions and blurred boundaries. While CNNs- and Transformer-based methods improve the classification of dominant large-area classes (e.g., grapes and vineyard), they still underperform on smaller or morphologically complex classes (e.g., broccoli green weeds 1/2 and lettuce romaine). Compared with other Mamba-based models, WD4SNet achieves more accurate and consistent boundary delineation for fine classes and better preserves the integrity of small-area regions, reaffirming its superiority in multiscale feature extraction and fine-grained classification. On the complex HU dataset [see Fig. 13], WD4SNet continues to exhibit strong performance and visual superiority. The method excels in identifying small or sparsely distributed objects (e.g., tennis courts, running tracks, and small parking lots), with prediction maps closely matching the ground truth regions. While conventional CNN- and Transformer-based methods in Fig. 13(c)–(h) show partial improvements in large-area classes (e.g., residential and commercial), they still struggle with small targets and intricate boundary regions. Other Mamba-based models in Fig. 13(i)–(m) approach the ground truth, but WD4SNet consistently provides the most accurate, detailed, and noise-free predictions, especially for challenging and minority classes. These qualitative results across all datasets confirm that WD4SNet not only achieves high OA but also ensures spatial consistency, boundary sharpness, and robust detection of both major and minority classes.

3) Visualization of Feature Classification: To further evaluate the discriminability of features learned by different methods, we visualize the feature distributions using t-distributed stochastic neighbor embedding (t-SNE) [53] for all benchmark datasets, as illustrated in Fig. 14. In each plot, different colors indicate different classes, and the

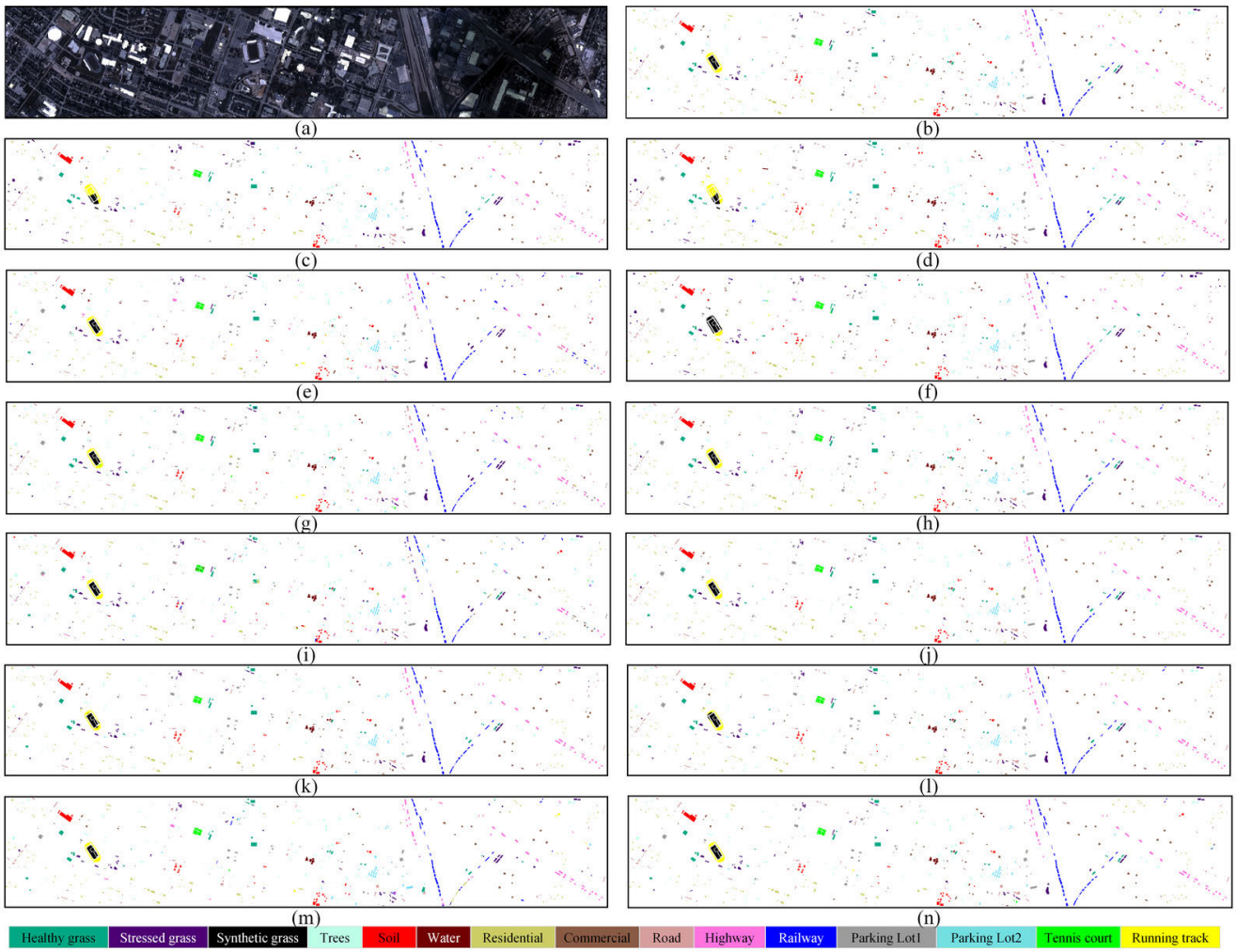


Fig. 13. Classification maps obtained by different methods on HU. (a) False-color image. (b) Ground truth. (c) SVM. (d) ASSMN. (e) 3D-CNN. (f) A2S2K. (g) SSFTT. (h) HiT. (i) SPRLT. (j) MASSF. (k) MIM. (l) DBMG. (m) FAHM. (n) WD4SNet (Ours).

clustering pattern provides insight into both intraclass compactness and interclass separability. Across all datasets, it is evident that the features extracted by WD4SNet exhibit the most compact intraclass clusters and the clearest interclass boundaries. Specifically, samples from the same class are projected into tightly grouped and well-separated clusters, indicating that WD4SNet effectively learns highly consistent intraclass representations and maximizes discrimination between different classes. This is particularly pronounced in challenging datasets, such as IP and HU, where other methods show significant class overlap and scattered sample distributions, while WD4SNet maintains clear and distinct clusters. Compared with traditional CNN- and Transformer-based approaches, WD4SNet consistently reduces intraclass variance and enhances interclass margins in the low-dimensional space. This suggests that the model captures more essential and robust spectral–spatial features, contributing to its superior classification performance observed in quantitative results. Notably, even for classes with limited samples or high spectral similarity, WD4SNet preserves cluster compactness and minimizes misclassification boundaries, underscoring its strong generalization ability.

F. Ablation Studies

To comprehensively validate the effectiveness of each proposed component, we conducted extensive ablation studies on the four benchmark datasets. The results are summarized in Tables IX–XII, and the analyses are as follows.

1) Effectiveness of the SSFB: We evaluated several variants of the SSFB to investigate the contributions of its components: 1) a spatial-only branch; 2) a spectral–spatial-only branch; 3) a variant without the adaptive gating mechanism (i.e., direct summation of the two branches); and 4) a comparison between standard convolution and weighted convolution within the block. As shown in Table IX, single-branch variants (spatial or spectral–spatial only) lead to a notable decrease in accuracy, underscoring the necessity of joint spectral–spatial modeling. The dual-branch design achieves significantly better results than either branch alone, confirming the complementarity of spatial and spectral–spatial information. Furthermore, the incorporation of the adaptive gating mechanism yields additional performance improvement, demonstrating its ability to dynamically fuse features and suppress redundancy.

Moreover, Table X demonstrates that using weighted convolution in place of standard convolution consistently enhances

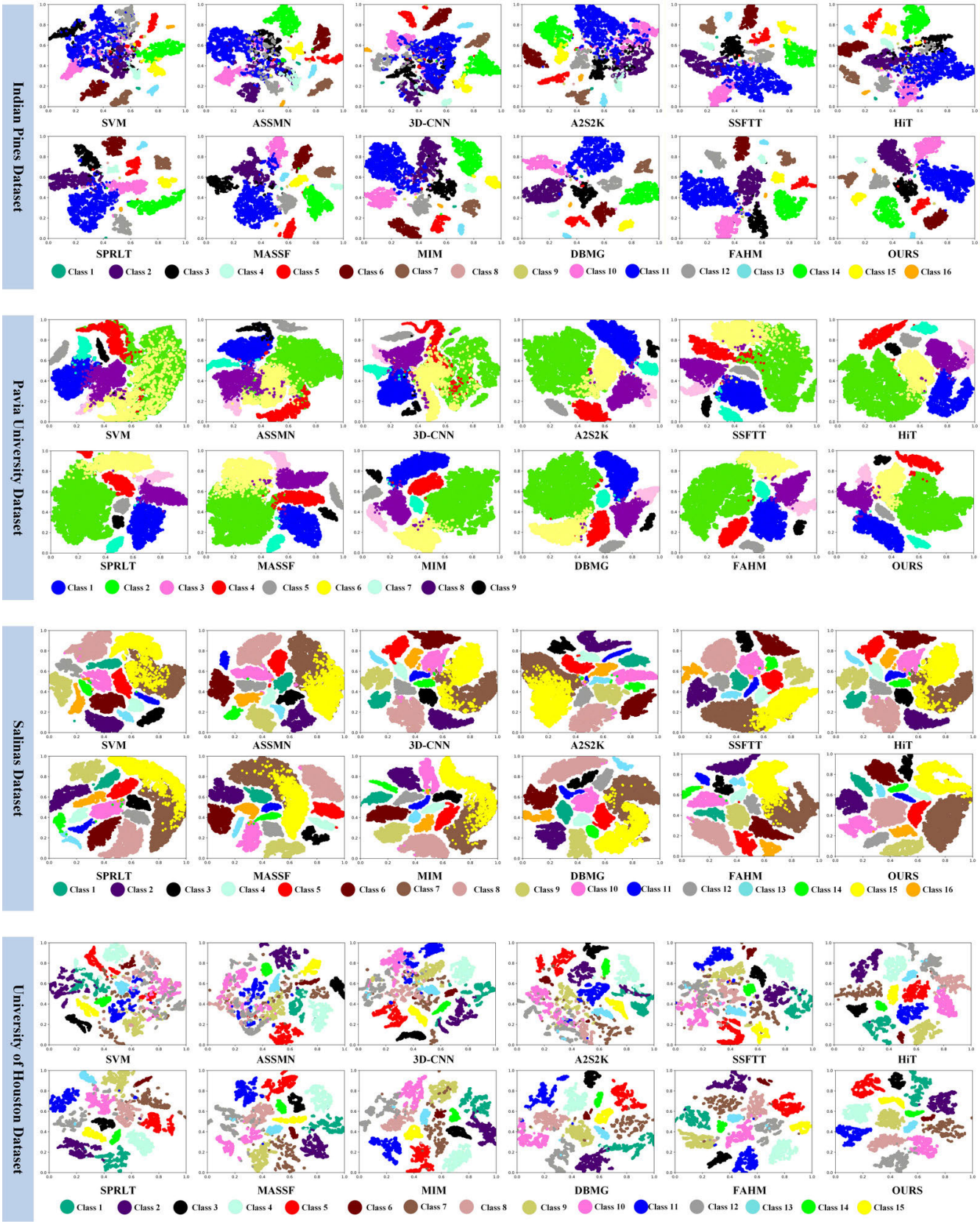


Fig. 14. Visualization of feature classification by different methods on the IP, PU, SA, and HU datasets.

OA, AA, and Kappa across all datasets. For example, on the IP dataset, OA, AA, and Kappa improved by 0.11%, 0.25%, and 0.43%, respectively.

2) Effectiveness of the S4M: To assess the contributions of each S4M submodule, we performed ablation by individually removing the SMSS, SASS, and CASFA. As reported in

TABLE IX
ABLATION STUDY OF THE SSFB ON FOUR DATASETS

Dataset	Metrics	Only Spatial	Only Spectral-Spatial	No Gating	**SSFB**
IP	OA(%)	70.10	88.46	94.89	**97.36**
	AA(%)	71.88	89.56	95.78	**98.40**
	$\kappa\times100$	70.66	88.95	94.99	**96.90**
PU	OA(%)	78.33	85.56	94.69	**98.47**
	AA(%)	79.89	86.71	94.12	**97.11**
	$\kappa\times100$	79.54	86.25	94.53	**97.92**
SA	OA(%)	84.33	89.23	96.35	**98.77**
	AA(%)	83.68	90.05	96.68	**98.78**
	$\kappa\times100$	84.22	90.19	97.85	**98.63**
HU	OA(%)	75.32	85.98	92.06	**96.40**
	AA(%)	76.75	86.88	93.36	**96.55**
	$\kappa\times100$	75.45	86.21	92.24	**96.09**

TABLE X
ABLATION STUDY OF S4M ON FOUR DATASETS

Dataset	SSFB		S4M			LFPD	OA(%)	AA(%)	Kappa ($\times100$)
	Conv	wConv	SMSS	SASS	CASFA				
IP	×	✓	×	×	×	✓	91.33	91.89	89.49
	×	✓	×	✓	✓	✓	93.89	93.56	93.06
	×	✓	✓	×	✓	✓	94.66	94.78	92.58
	×	✓	✓	✓	×	✓	95.45	96.04	93.48
	×	✓	✓	✓	✓	×	97.23	96.66	96.03
	✓	×	✓	✓	✓	✓	97.25	98.15	96.47
	×	✓	✓	✓	✓	✓	**97.36**	**98.40**	**96.90**
PU	×	✓	×	×	×	✓	92.52	93.17	92.37
	×	✓	×	✓	✓	✓	93.30	94.09	93.12
	×	✓	✓	×	✓	✓	92.76	93.89	92.56
	×	✓	✓	✓	×	✓	94.13	94.45	91.65
	×	✓	✓	✓	✓	×	94.54	94.95	93.36
	✓	×	✓	✓	✓	✓	98.16	97.10	97.88
	×	✓	✓	✓	✓	✓	**98.47**	**97.11**	**97.92**
SA	×	✓	×	×	×	✓	95.02	95.88	94.97
	×	✓	×	✓	✓	✓	95.36	96.45	95.22
	×	✓	✓	×	✓	✓	95.55	96.77	95.48
	×	✓	✓	✓	×	✓	96.28	97.56	96.18
	×	✓	✓	✓	✓	×	96.44	97.36	96.12
	✓	×	✓	✓	✓	✓	98.54	98.49	98.36
	×	✓	✓	✓	✓	✓	**98.77**	**98.78**	**98.63**
HU	×	✓	×	×	×	✓	92.00	92.38	91.88
	×	✓	×	✓	✓	✓	92.77	93.06	92.64
	×	✓	✓	×	✓	✓	92.63	92.95	92.55
	×	✓	✓	✓	×	✓	94.89	96.10	94.78
	×	✓	✓	✓	✓	×	95.13	95.25	94.95
	✓	×	✓	✓	✓	✓	95.99	96.06	95.84
	×	✓	✓	✓	✓	✓	**96.40**	**96.55**	**96.09**

Table X, eliminating either SMSS or SASS leads to substantial performance degradation. This is because the shallow features fused by SSFB alone are insufficient to capture deep, discriminative spectral–spatial patterns. SMSS and SASS enable effective modeling of multiscale semantic information along the spectral and spatial axes, respectively, thus providing robust and noise-resistant features for classification. Excluding CASFA also causes a noticeable decrease in accuracy, indicating its importance in aligning and integrating spectral–spatial features across axes. The inclusion of all S4M components yields OA improvements of 6.03% (IP), 5.95% (PU), 3.75% (SA), and 4.40% (HU) compared to the most simplified ablated variants, highlighting the indispensable role of each component.

To further investigate the influence of different wavelet bases, we compare Haar, Daubechies, and Symlets in the proposed wavelet-guided modules. As shown in Table XII, Haar achieves the best or comparable performance across the four datasets. Daubechies and Symlets provide smoother basis functions, but their longer filter support may introduce more smoothing and slightly higher computational cost, which is less favorable for preserving fine spatial boundaries and local spectral variations in HSI classification. Therefore, Haar is adopted as the default wavelet basis in WD4SNet.

3) Effectiveness of the LFPD: The LFPD is designed to facilitate effective multiscale feature fusion in the decoding phase. To examine its impact, we replaced LFPD with several alternative decoding strategies: 1) simple upsampling and concatenation (bilinear + conv); 2) direct concatenation of upsampled multiscale features followed by convolution (without progressive top-down fusion or depthwise convolution smoothing); and 3) replacing ASPP-lite with standard ASPP. As summarized in Table XI, integrating LFPD leads to the highest accuracy on all datasets, with OA improvements

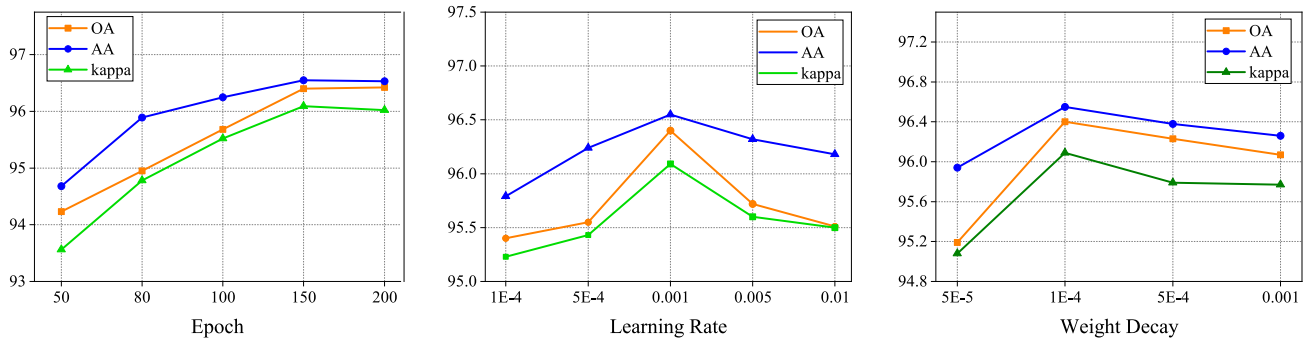


Fig. 15. Impact of training parameters, including epoch, learning rate, and weight decay.

TABLE XI
ABLATION STUDY OF LFPD ON FOUR DATASETS

Method	Evaluation	IP	PU	SA	HU
Bilinear+Conv	OA(%)	96.42	97.71	98.18	95.29
	AA(%)	97.31	96.15	98.15	95.39
	Kappa×100	95.93	96.95	97.97	94.91
	Params(M)	1.45	1.87	2.15	1.59
	FLOPs(G)	0.058	0.062	0.065	0.061
Concat+Conv	OA(%)	96.81	98.01	98.35	95.71
	AA(%)	97.73	96.45	98.35	95.82
	Kappa×100	96.37	97.33	98.19	95.36
	Params(M)	1.33	1.72	2.09	1.58
	FLOPs(G)	0.055	0.057	0.059	0.058
ASPP	OA(%)	97.14	98.25	98.59	96.08
	AA(%)	98.06	96.83	98.56	96.18
	Kappa×100	96.66	97.64	98.43	95.71
	Params(M)	1.51	1.95	2.38	1.80
	FLOPs(G)	0.063	0.065	0.068	0.067
LFPD	OA(%)	**97.36**	**98.47**	**98.77**	**96.40**
	AA(%)	**98.40**	**97.11**	**98.78**	**96.55**
	Kappa×100	**96.90**	**97.92**	**98.63**	**96.09**
	Params(M)	**1.21**	**1.56**	**1.90**	**1.44**
	FLOPs(G)	**0.049**	**0.051**	**0.053**	**0.052**

TABLE XII
IMPACT OF WAVELET BASIS ON FOUR DATASETS

Wavelet basis	Evaluation	IP	PU	SA	HU
Daubechies/db2	OA(%)	97.04	98.21	98.48	96.09
	AA(%)	98.06	96.78	98.51	96.22
	Kappa×100	96.55	97.60	98.31	95.76
Symlets/sym2	OA(%)	97.12	98.29	98.54	96.17
	AA(%)	98.18	96.86	98.59	96.31
	Kappa×100	96.63	97.69	98.37	95.83
Haar/db1	OA(%)	**97.36**	**98.47**	**98.77**	**96.40**
	AA(%)	**98.40**	**97.11**	**98.78**	**96.55**
	Kappa×100	**96.90**	**97.92**	**98.63**	**96.09**

of 0.22% (IP), 0.22% (PU), 0.18% (SA), and 0.32% (HU) compared to the best baseline. LFPD consistently outperforms other approaches by a significant margin, especially in terms of spatial consistency and detail preservation. These results confirm the effectiveness of LFPD in achieving robust multiscale feature integration and enhancing final classification quality.

G. Parameter Analysis

To better understand the sensitivity and robustness of WD4SNet, we conducted a series of experiments to analyze the impact of key hyperparameters and architectural settings on classification performance across four benchmark datasets.

1) Impact of Training Hyperparameters: We investigated the influence of three critical training parameters, number of epochs, learning rate, and weight decay, using the HU dataset as a representative example [see Fig. 15]. Optimal performance is achieved at 150 epochs; further increases lead to marginal performance decrease and unnecessary computational overhead, likely due to overfitting. Regarding learning rate, too small a value slows convergence, while an excessively large value induces oscillation or even divergence. The best OA, AA, and Kappa are obtained at a learning rate of 0.001. For weight decay, a value of 0.0001 strikes the right balance between model complexity and generalization, effectively mitigating overfitting. Overall, the combination of 150 epochs, a learning rate of 0.001, and weight decay of 0.0001 yields optimal tradeoffs between accuracy, efficiency, and robustness.

TABLE XIII
IMPACT OF MAMBA EMBEDDING SIZE IN FOUR DATASETS

$\{C_1, C_2, C_3\}$	Evaluation	IP	PU	SA	HU
$\{144, 272, 368\}$	OA(%)	96.52	98.05	98.31	95.48
	AA(%)	97.68	96.54	98.29	95.64
	Kappa×100	96.03	97.40	98.12	95.10
$\{176, 368, 448\}$	OA(%)	97.01	98.28	98.58	96.03
	AA(%)	98.06	96.88	98.56	96.18
	Kappa×100	96.60	97.66	98.42	95.69
$\{192, 384, 448\}$	OA(%)	97.28	98.41	98.69	96.27
	AA(%)	98.31	97.02	98.67	96.39
	Kappa×100	96.81	97.84	98.55	95.94
$\{200, 376, 448\}$	OA(%)	**97.36**	**98.47**	**98.77**	**96.40**
	AA(%)	**98.40**	**97.11**	**98.78**	**96.55**
	Kappa×100	**96.90**	**97.92**	**98.63**	**96.09**

2) Impact of Mamba Embedding Size: We further analyzed the effect of embedding dimensionality on the network's feature representation capacity. Guided by Mobile-Mamba configurations [54], we tested four combinations of stage-wise embedding sizes: $\{144, 272, 368\}$, $\{176, 368, 448\}$, $\{192, 384, 448\}$, and $\{200, 376, 448\}$, with all other settings held constant. As shown in Table XIII, increasing the channel width consistently improves OA, AA, and Kappa across all datasets, reflecting enhanced discriminative power for spectral–spatial features. Notably, the configuration $\{200, 376, 448\}$ achieves the highest performance, with OA reaching 98.77% on SA and above 96% on the other datasets. This demonstrates that

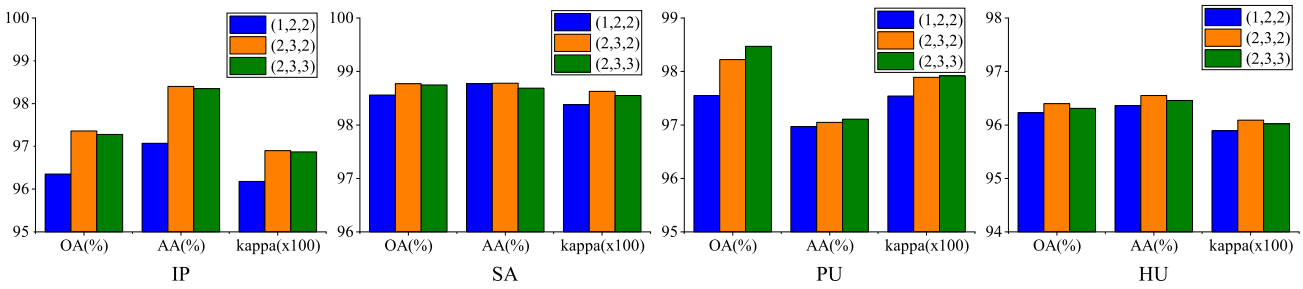


Fig. 16. Impact of Mamba depth on different datasets, including IP, PU, SA, and HU.

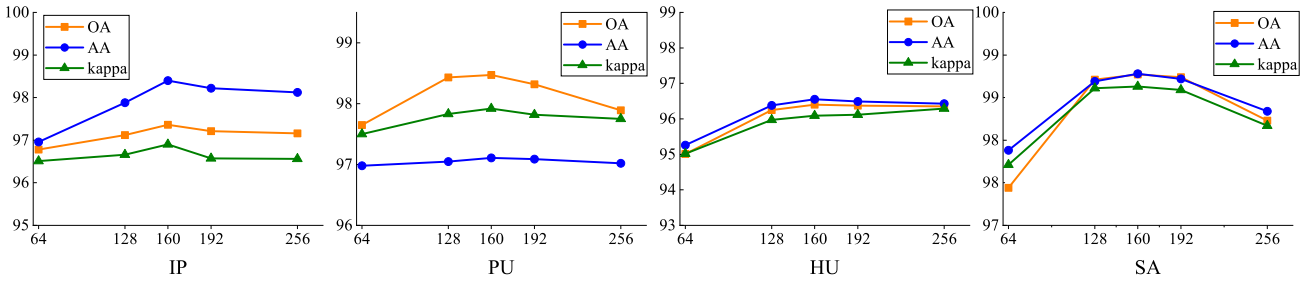


Fig. 17. Impact of decoder dimension on different datasets, including IP, PU, SA, and HU.

TABLE XIV
COMPUTATIONAL PARAMETER ANALYSIS OF VARIOUS COMPARISON METHODS IN TERMS OF PARAMETERS, FLOPS, AND RUNNING TIME ON FOUR DATASETS

Datasets	Metric	SVM	ASSMN	3D-CNN	A2S2K	SSFTT	HiT	SPRLT	MASSF	MIM	DBMG	FAHM	WD4SNet
IP	Training time (s)	0.20	18.24	22.12	284.20	27.19	134.52	1.944	34.88	692.44	69.84	53.69	198.4
	Testing time (s)	1.20	21.83	23.83	2.97	0.48	7.25	9.20	1.34	19.25	1.91	2.08	1.42
	Parameters (M)	–	2.845	3.139	5.64	1.53	59.286	0.846	1.459	1.171	1.458	1.077	1.21
	FLOPs (G)	–	0.015	0.111	0.123	0.021	0.427	0.035	0.052	0.051	0.059	0.082	0.049
PU	Training time (s)	0.66	12.89	3.5	460.49	105.14	945.65	1.257	112.75	3600.54	274.56	289.01	269.45
	Testing time (s)	16.11	56.54	4.74	16.30	5.09	125.67	65.76	9.56	192.42	18.10	18.92	1.23
	Parameters (M)	–	2.969	2.918	2.21	1.53	54.505	0.833	0.927	1.169	1.735	0.681	1.56
	FLOPs (G)	–	0.014	0.102	0.062	0.020	0.329	0.035	0.135	0.051	0.051	0.049	0.051
SA	Training time (s)	0.52	1.82	82.39	336.83	101.16	151.71	0.969	18.35	3699.66	385.56	75.09	168.44
	Testing time (s)	0.98	14.57	4.72	18.82	5.45	34.35	20.01	2.06	203.12	59.38	29.92	1.02
	Parameters (M)	–	2.874	3.203	4.80	1.53	53.382	0.839	0.153	1.161	1.562	1.485	1.90
	FLOPs (G)	–	0.015	0.118	0.120	0.020	0.306	0.035	0.004	0.051	0.062	0.095	0.053
HU	Training time (s)	3.64	8.76	2.36	364.31	68.39	179.12	0.207	46.60	2373.52	255.98	75.49	197.9
	Testing time (s)	4.10	37.00	2.43	30.11	17.47	96.37	11.21	16.95	614.89	56.62	9.89	1.93
	Parameters (M)	–	2.774	2.1036	2.83	1.53	64.797	0.845	0.248	1.168	1.408	2.54	1.44
	FLOPs (G)	–	0.014	0.098	0.059	0.021	0.538	0.035	0.024	0.050	0.062	0.10	0.052

moderately larger embedding sizes strengthen the model's ability to capture fine-grained spectral patterns and long-range dependencies, without incurring excessive computational cost.

3) Impact of Mamba Depth: The network depth directly affects the abstraction and modeling capacity of spectral–spatial features. We evaluated three stacking configurations: $(1, 2, 2)$, $(2, 3, 2)$, and $(2, 3, 3)$, corresponding to shallow, medium, and deep architectures, respectively. Fig. 16 presents the results. Increasing depth generally enhances performance, especially on more challenging datasets such as IP and PU, where deeper stages facilitate the extraction of complex spectral–spatial relationships. For example, on the IP dataset, OA rises from 96.35% (shallow) to 98.66% (deep). On the SA and HU datasets, however, further increasing depth brings diminishing or negative returns, possibly due to over-smoothing or overfitting, suggesting that optimal depth should be tailored to the complexity of each dataset.

4) Impact of Decoder Channel Width: We assessed the sensitivity of the LFPD to its channel width parameter, which determines the model's capacity for multiscale feature fusion and semantic reconstruction. As depicted in Fig. 17, increasing the decoder dimension from 64 to 128 or 160 channels leads to a remarkable improvement in OA, AA, and Kappa

across all datasets. However, further increases yield marginal gains, suggesting a sweet spot between fusion capability and computational efficiency. Notably, too narrow a decoder (e.g., 64 channels) results in underfitting and loss of spectral–spatial detail, especially in complex scenes.

H. Computational Complexity Analysis

Table XIV presents a comprehensive comparison of training time, inference time, and parameter count for representative models across the four benchmark datasets. Traditional machine learning methods, such as SVM and ASSMN, possess the lowest parameter counts and fastest inference speeds due to their shallow architectures. However, their limited capacity to capture intricate spectral–spatial correlations results in suboptimal OA, restricting their applicability in complex hyperspectral tasks. CNN-based methods, leveraging local convolutions and, in some cases, 3-D kernels, significantly enhance feature extraction but at the cost of larger parameter sizes and increased inference latency. Particularly, 3D-CNNs incur substantial computational burdens, as reflected in their longer training and testing times in Table XIV. Transformer-based models demonstrate strong global modeling capability and superior feature representation, contributing to higher classification performance. However, the quadratic computational complexity of multihead self-attention with respect to input size leads to considerable memory and runtime overhead, especially for high-dimensional hyperspectral data. This results in an unfavorable tradeoff between accuracy and efficiency, which can be a bottleneck for large-scale or real-time remote sensing applications. In contrast, the proposed WD4SNet achieves a favorable balance between accuracy and efficiency. By incorporating the linear-time state space mechanism, along with the efficient spectral–spatial fusion enabled by the SSFB and Mamba architecture, WD4SNet achieves a favorable accuracy–efficiency tradeoff and uses far fewer parameters than heavy Transformer models such as HiT. In addition, the lightweight decoder further streamlines computation by minimizing redundant operations while maintaining high-fidelity spatial detail reconstruction.

V. Conclusion

Although WD4SNet achieves superior overall performance across the four benchmark datasets, it does not consistently obtain the highest accuracy for every individual category. This observation indicates that several challenging classes remain difficult to distinguish, especially those with highly similar spectral signatures, limited training samples, or complex spatial structures. For example, in agricultural scenes, different crop categories at similar growth stages often exhibit overlapping spectral responses, which increases interclass confusion. In complex urban scenes, mixed pixels, irregular boundaries, and fine-scale structures further complicate accurate classification. Although WD4SNet enhances spectral–spatial representation through wavelet-driven multiscale state-space modeling and cross-axis feature alignment, some subtle discriminative cues may still be weakened during hierarchical feature abstraction and reconstruction.

These limitations suggest several promising directions for future improvement. First, introducing contrastive learning or prototype-based class separation may further enlarge interclass margins for spectrally similar categories. Second, boundary-aware refinement strategies or high-resolution detail recovery modules could improve the recognition of classes with complex structures and ambiguous edges. Third, more effective learning schemes for hard samples and minority classes may help reduce the residual confusion in difficult categories. Therefore, while the proposed WD4SNet demonstrates strong accuracy, efficiency, and scalability, further enhancing fine-grained class discrimination remains an important direction for future research.

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